ECE131: Basic Electrical and Electronics Engineering

Comprehensive University Question Bank & Solved Problems

LPU Engineering Academic Faculty & Editorial Board

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Chapter 1

Unit 1: Fundamentals of D.C. Circuits

1.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. The SI unit of electrical resistance is the: [Topic: resistance | Easy]
(a) Ohm (Ω)
(b) Farad (F)
(c) Ampere (A)
(d) Volt (V)

Q2. The resistance of a conductor is directly proportional to its: [Topic: resistance | Easy]
(a) Current through it
(b) Length
(c) Temperature coefficient only
(d) Cross-sectional area

Q3. According to Ohm's law, the current through a resistor is given by: [Topic: ohm's law | Easy]
(a) $I = V + R$
(b) $I = V \cdot R$
(c) $I = V\ R$
(d) $I = R\ V$

Q4. A 10 Ω resistor carries a current of 2 A . The voltage across it is: [Topic: ohm's law | Easy]
(a) 20 V
(b) 0.2 V
(c) 12 V
(d) 5 V

Q5. Voltage is best described as the: [Topic: voltage | Easy]
(a) Opposition to current flow
(b) Stored electric charge
(c) Rate of flow of charge
(d) Potential difference between two points

Q6. Electric current is defined as the rate of flow of: [Topic: current | Easy]
(a) Power
(b) Voltage
(c) Energy
(d) Charge

Q7. The SI unit of capacitance is the: [Topic: capacitance | Easy]
(a) Ohm (Ω)
(b) Coulomb (C)
(c) Henry (H)

(d) Farad (F)

Q8. The charge stored on a capacitor is given by: [Topic: capacitance | Easy]

(a) Q = C + V
(b) Q = C V
(c) Q = V C
(d) Q = C V

Q9. The SI unit of inductance is the: [Topic: inductance | Easy]

(a) Weber (Wb)
(b) Farad (F)
(c) Tesla (T)
(d) Henry (H)

Q10. Electrical power dissipated in a resistor can be expressed as: [Topic: power and energy concepts | Easy]

(a) P = V I
(b) P = V I
(c) P = V + I
(d) P = I V

Q11. The unit of electrical energy is the: [Topic: power and energy concepts | Easy]

(a) Ampere (A)
(b) Joule (J)
(c) Volt (V)
(d) Watt (W)

Q12. Kirchhoff's Current Law (KCL) is based on the conservation of: [Topic: kirchhoff's laws | Easy]

(a) Energy
(b) Voltage
(c) Power
(d) Charge

Q13. Kirchhoff's Voltage Law (KVL) states that the algebraic sum of voltages around a closed loop is: [Topic: kirchhoff's laws | Easy]

(a) Equal to the total current
(b) Zero
(c) Infinite
(d) Equal to the source voltage

Q14. Three resistors of 2 Ω , 3 Ω , and 5 Ω are connected in series. The total resistance is: [Topic: intuitive method of circuit analysis- series and parallel simplification | Easy]

(a) 10 Ω
(b) 30 Ω
(c) 3.33 Ω
(d) 0.97 Ω

Q15. Two 6 Ω resistors connected in parallel give a total resistance of: [Topic: intuitive method of circuit analysis- series and parallel simplification | Easy]

(a) 3 Ω
(b) 36 Ω
(c) 12 Ω
(d) 6 Ω

Q16. In a series circuit, the voltage division rule states that the voltage across a resistor is proportional to: [Topic: voltage division rule | Easy]

(a) The source current only
(b) The reciprocal of its resistance

(c) The total current squared
(d) Its own resistance

Q17. In a parallel circuit, the current division rule shows that more current flows through the branch with: [Topic: current division rule | Easy]
(a) Higher resistance
(b) Lower resistance
(c) Longer wire length
(d) Larger inductance

Q18. A source whose output depends on a voltage or current elsewhere in the circuit is called a: [Topic: introduction to dependent and independent sources | Easy]
(a) Ideal source
(b) Short circuit
(c) Dependent source
(d) Independent source

Q19. According to the maximum power transfer theorem, maximum power is delivered to the load when the load resistance equals the: [Topic: maximum power transfer theorem | Easy]
(a) Twice the source resistance
(b) Total circuit current
(c) Source voltage
(d) Source (Thevenin) resistance

Q20. The superposition theorem is applicable only to circuits that are: [Topic: network theorems-superposition theorem | Easy]
(a) Linear
(b) Purely capacitive
(c) Nonlinear
(d) Open circuited

Q21. A resistor carries a current of 0.5 A when connected across a 10 V supply. If the voltage is increased to 25 V , what current flows through the same resistor? [Topic: ohm's law | Medium]
(a) 0.5 A
(b) 0.2 A
(c) 2.5 A
(d) 1.25 A

Q22. Three resistors of 6 Ω , 3 Ω , and 2 Ω are connected in parallel. What is the equivalent resistance? [Topic: resistance | Medium]
(a) 3.67 Ω
(b) 1 Ω
(c) 11 Ω
(d) 0.5 Ω

Q23. A 100 Ω resistor dissipates 4 W of power. What is the current flowing through it? [Topic: power and energy concepts | Medium]
(a) 0.04 A
(b) 0.2 A
(c) 0.4 A
(d) 2 A

Q24. Two capacitors of 4 μ F and 12 μ F are connected in series. What is the equivalent capacitance? [Topic: capacitance | Medium]
(a) 16 μ F
(b) 3 μ F
(c) 8 μ F
(d) 48 μ F

Q25. The voltage across an inductor of 0.5 H when the current changes at a rate of 4 A / s is: [Topic: inductance | Medium]

(a) 2 V
(b) 8 V
(c) 4.5 V
(d) 0.125 V

Q26. In a series circuit, a 12 V source is connected across resistors of 3 Ω and 9 Ω . What is the voltage across the 9 Ω resistor? [Topic: voltage division rule | Medium]

(a) 9 V
(b) 6 V
(c) 3 V
(d) 4 V

Q27. A total current of 6 A enters two parallel resistors of 2 Ω and 4 Ω . What current flows through the 4 Ω resistor? [Topic: current division rule | Medium]

(a) 4 A
(b) 3 A
(c) 1.5 A
(d) 2 A

Q28. At a node, three currents enter: 5 A , 3 A , and 2 A , while one current leaves as 4 A . What is the magnitude of the second leaving current? [Topic: kirchhoff's laws | Medium]

(a) 2 A
(b) 4 A
(c) 10 A
(d) 6 A

Q29. In a closed loop, the sum of EMF sources is 20 V . If two resistors drop 8 V and 7 V , what is the voltage drop across the third resistor? [Topic: kirchhoff's laws | Medium]

(a) 15 V
(b) 5 V
(c) 12 V
(d) 35 V

Q30. A 6 Ω resistor is in series with a parallel combination of two 4 Ω resistors. What is the total resistance? [Topic: intuitive method of circuit analysis- series and parallel simplification | Medium]

(a) 10 Ω
(b) 8 Ω
(c) 14 Ω
(d) 2 Ω

Q31. Three equal resistors of 9 Ω are connected in delta. What is the resistance of each arm of the equivalent star network? [Topic: star–delta transformation | Medium]

(a) 3 Ω
(b) 27 Ω
(c) 9 Ω
(d) 6 Ω

Q32. Which of the following best describes a dependent source? [Topic: introduction to dependent and independent sources | Medium]

(a) Its value is fixed regardless of circuit conditions
(b) It can only absorb power, never deliver it
(c) It supplies only alternating current
(d) Its value is controlled by a voltage or current elsewhere in the circuit

Q33. For a planar circuit with 4 branches and 3 nodes, how many independent mesh equations are required? [Topic: mesh and nodal analysis | Medium]

(a) 1
(b) 3
(c) 4
(d) 2

Q34. In nodal analysis, how many node equations are needed for a circuit having 5 nodes (including the reference node)? [Topic: mesh and nodal analysis | Medium]

(a) 6
(b) 4
(c) 3
(d) 5

Q35. A network has an open-circuit voltage of 12 V and delivers a short-circuit current of 3 A at its terminals. What is the Thevenin resistance? [Topic: Thevenin's theorem | Medium]

(a) 9 Ω
(b) 0.25 Ω
(c) 36 Ω
(d) 4 Ω

Q36. A Thevenin equivalent has V t h = 20 V and R t h = 5 Ω . What is the Norton current of the equivalent circuit? [Topic: Norton's theorem | Medium]

(a) 100 A
(b) 4 A
(c) 0.25 A
(d) 15 A

Q37. A source has Thevenin resistance 8 Ω and open-circuit voltage 16 V . What is the maximum power delivered to a matched load? [Topic: maximum power transfer theorem | Medium]

(a) 32 W
(b) 4 W
(c) 16 W
(d) 8 W

Q38. For maximum power transfer to a resistive load in a DC circuit, the load resistance should be: [Topic: maximum power transfer theorem | Medium]

(a) Zero
(b) Infinite
(c) Equal to the Thevenin resistance of the source
(d) Much larger than the source resistance

Q39. While applying the superposition theorem, an independent voltage source that is not being considered must be: [Topic: network theorems- superposition theorem | Medium]

(a) Doubled in value
(b) Left unchanged
(c) Replaced by an open circuit
(d) Replaced by a short circuit

Q40. A charge of 120 C flows through a conductor in 2 minutes. What is the average current? [Topic: current | Medium]

(a) 2 A
(b) 1 A
(c) 60 A
(d) 240 A

Q41. A balanced delta network has three equal resistors of R Δ = 18 Ω each. If this is converted to an equivalent star network, what is the value of each star resistor? [Topic: star–delta transformation | Hard]

(a) 54 Ω

(b) 6 Ω
(c) 9 Ω
(d) 3 Ω

Q42. A network has an open-circuit voltage of 24 V and delivers a short-circuit current of 4 A at its terminals. A load resistor R L = 8 Ω is connected. What current flows through R L ? [Topic: Thevenin's theorem | Hard]

(a) 1 A
(b) 3 A
(c) 2 A
(d) 1.71 A

Q43. A source with Thevenin voltage 20 V and internal resistance 5 Ω drives a variable load. What is the maximum power that can be delivered to the load? [Topic: maximum power transfer theorem | Hard]

(a) 10 W
(b) 40 W
(c) 80 W
(d) 20 W

Q44. In a linear circuit, a 10 V source alone produces 2 A in a branch, and a 5 A source alone produces - 1 A in the same branch. With both sources active, what is the branch current? [Topic: network theorems- superposition theorem | Hard]

(a) 3 A
(b) 1 A
(c) - 3 A
(d) 2 A

Q45. A two-node circuit (plus reference) has node equations: 3 V 1 - V 2 = 10 and - V 1 + 2 V 2 = 5 (conductances in S, currents in A). What is V 1 ? [Topic: mesh and nodal analysis | Hard]

(a) 2.5 V
(b) 4 V
(c) 5 V
(d) 3 V

Q46. A total current of 12 A enters two parallel resistors R 1 = 4 Ω and R 2 = 12 Ω . What current flows through R 1 ? [Topic: current division rule | Hard]

(a) 3 A
(b) 9 A
(c) 6 A
(d) 8 A

Q47. Three resistors 2 Ω , 3 Ω , and 5 Ω are in series across a 20 V source. What voltage appears across the 3 Ω resistor? [Topic: voltage division rule | Hard]

(a) 3 V
(b) 4 V
(c) 10 V
(d) 6 V

Q48. At a node, three currents meet: I 1 = 5 A entering, I 2 = 3 A leaving, and I 3 unknown. If KCL holds and I 3 leaves the node, what is I 3 ? [Topic: kirchhoff's laws | Hard]

(a) 2 A
(b) 3 A
(c) - 2 A
(d) 8 A

Q49. A Norton equivalent has I N = 6 A and R N = 4 Ω . A load of 4 Ω is connected. What power is dissipated in the load? [Topic: Norton's theorem | Hard]

(a) 18 W
(b) 144 W
(c) 36 W
(d) 72 W

Q50. Two capacitors C 1 = 6 μ F and C 2 = 3 μ F are connected in series across 30 V . What is the charge stored on each capacitor? [Topic: capacitance | Hard]

(a) 180 μ C
(b) 60 μ C
(c) 90 μ C
(d) 30 μ C

Q51. The current through a 0.5 H inductor changes from 2 A to 6 A in 0.1 s . What is the magnitude of the induced EMF? [Topic: inductance | Hard]

(a) 5 V
(b) 40 V
(c) 20 V
(d) 10 V

Q52. A 100 W device operates for 5 hours daily. What is the energy consumed in one week in kWh? [Topic: power and energy concepts | Hard]

(a) 0.5 kWh
(b) 35 kWh
(c) 3.5 kWh
(d) 7 kWh

Q53. A wire of resistance R is stretched uniformly to twice its original length (volume constant). What is its new resistance? [Topic: resistance | Hard]

(a) R /2
(b) R /4
(c) 2 R
(d) 4 R

Q54. A resistor dissipates 50 W when carrying 5 A . What voltage is across it? [Topic: ohm's law | Hard]

(a) 250 V
(b) 10 V
(c) 55 V
(d) 2 V

Q55. Two 6 Ω resistors in parallel are placed in series with a 4 Ω resistor, and this combination is in parallel with a 7 Ω resistor. What is the total resistance? [Topic: intuitive method of circuit analysis-series and parallel simplification | Hard]

(a) 3.5 Ω
(b) 7 Ω
(c) 10 Ω
(d) 2.33 Ω

Q56. A circuit contains a current-controlled voltage source V s = 4 I x where I x is the current through a 2 Ω resistor carrying 3 A . What is the source voltage? [Topic: introduction to dependent and independent sources | Hard]

(a) 12 V
(b) 6 V
(c) 8 V
(d) 24 V

Q57. A circuit has 4 nodes and 6 branches. How many independent KVL (loop) equations are needed for complete analysis? [Topic: basic method of circuit analysis | Hard]

(a) 2
(b) 3
(c) 6
(d) 4

Q58. Moving a charge of 4 C between two points requires 48 J of work. What is the potential difference between the points? [Topic: voltage | Hard]

(a) 192 V
(b) 44 V
(c) 0.083 V
(d) 12 V

Q59. A charge of 600 C passes through a conductor in 5 minutes. What is the average current? [Topic: current | Hard]

(a) 10 A
(b) 2 A
(c) 0.5 A
(d) 120 A

Q60. A two-mesh circuit yields: 10 = 6 I 1 - 2 I 2 and 0 = - 2 I 1 + 5 I 2 . What is the mesh current I 1 ? [Topic: mesh and nodal analysis | Hard]

(a) 1.67 A
(b) 0.77 A
(c) 1.92 A
(d) 2.5 A

1.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 1 **[Subjective & Numerical Evaluation]**

1 In the circuit shown below, determine the branch currents I_1, I_2, and I_3 using Mesh Current Analysis. Also determine the Thevenin equivalent circuit across terminals $A-B$ connected across the $10\,\Omega$ load resistor, and find the maximum power transferable to R_L.

Circuit Parameters: Independent voltage sources $V_1 = 36$ V, $V_2 = 12$ V; Resistors $R_1 = 6\,\Omega$, $R_2 = 3\,\Omega$, $R_3 = 6\,\Omega$, Load $R_L = 10\,\Omega$.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Mesh Formulation and Matrix Setup
Define clockwise mesh currents I_a in Loop 1 (left loop) and I_b in Loop 2 (right loop):

$$\text{Loop 1: } (R_1+R_2)I_a - R_2 I_b = V_1 \implies 9I_a - 3I_b = 36$$
$$\text{Loop 2: } -R_2 I_a + (R_2+R_3)I_b = -V_2 \implies -3I_a + 9I_b = -12$$

Step 2: Solving for Mesh Currents via Cramer's Rule
The determinant of the impedance matrix is $\Delta = (9)(9) - (-3)(-3) = 81 - 9 = 72\,\Omega^2$.

$$I_a = \frac{\begin{vmatrix} 36 & -3 \\ -12 & 9 \end{vmatrix}}{72} = \frac{(36)(9) - (-12)(-3)}{72} = \frac{324 - 36}{72} = \frac{288}{72} = 4.0 \text{ A}$$
$$I_b = \frac{\begin{vmatrix} 9 & 36 \\ -3 & -12 \end{vmatrix}}{72} = \frac{(9)(-12) - (-3)(36)}{72} = \frac{-108 + 108}{72} = 0.0 \text{ A}$$

Branch currents are therefore: $I_{R1} = I_a = 4.0$ A, $I_{R2} = I_a - I_b = 4.0$ A, and $I_{R3} = I_b = 0.0$ A.

Step 3: Thevenin Equivalent (V_{th}, R_{th} across terminals of R_3)
Open-circuiting R_3, the open circuit voltage across terminals $A-B$ is governed by the voltage divider of V_1 across R_2 minus V_2:

$$V_{th} = V_1 \frac{R_2}{R_1+R_2} - V_2 = 36 \times \frac{3}{9} - 12 = 12 - 12 = 0 \text{ V}$$

Deactivating independent sources (short-circuiting V_1 and V_2):

$$R_{th} = (R_1 \parallel R_2) = \frac{6 \times 3}{6+3} = 2.0\,\Omega$$

For Maximum Power Transfer to load R_L, we require $R_L = R_{th} = 2.0\,\Omega$.

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

Grading Breakdown [Total: 10 Marks]:

- Correct KVL mesh equations with clear sign conventions: **3 Marks**
- Matrix inversion / determinant computation of branch currents: **3 Marks**
- Thevenin resistance R_{th} calculation with deactivated sources: **2 Marks**
- Maximum power theorem condition and final calculation: **2 Marks**

Chapter 2

Unit 2: Fundamentals of A.C. Circuits

2.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. An alternating current is one whose magnitude and direction: [Topic: alternating current and voltage | Easy]

(a) Remain constant with time
(b) Change periodically with time
(c) Only change in magnitude but not direction
(d) Only change in direction but not magnitude

Q2. The standard waveform assumed for an AC supply in analysis is: [Topic: alternating current and voltage | Easy]

(a) Triangular
(b) Sinusoidal
(c) Sawtooth
(d) Square

Q3. In AC circuit notation, lowercase letters such as i and v generally represent: [Topic: concept of notations (i, v, I, V) | Easy]

(a) Instantaneous values
(b) Maximum values
(c) RMS values
(d) Average values

Q4. Uppercase symbols I and V in AC analysis most commonly represent: [Topic: concept of notations (i, v, I, V) | Easy]

(a) Angular frequency
(b) Peak-to-peak values
(c) RMS (effective) values
(d) Instantaneous values

Q5. The amplitude of an AC waveform refers to its: [Topic: definitions of amplitude, phase, phase difference | Easy]

(a) Average value
(b) RMS value
(c) Time period
(d) Maximum (peak) value

Q6. Phase difference between two AC quantities is usually measured in: [Topic: definitions of amplitude, phase, phase difference | Easy]

(a) Amperes
(b) Degrees or radians
(c) Volts

(d) Seconds only

Q7. Two AC waveforms that reach their peaks at the same instant are said to be: [Topic: definitions of amplitude, phase, phase difference | Easy]

(a) In phase
(b) Out of phase by 18 0 °
(c) In quadrature
(d) Out of phase by 9 0 °

Q8. For a sinusoidal waveform of peak value V m , the RMS value is: [Topic: RMS value and average value of an AC signal | Easy]

(a) 2 V m
(b) 2 V m
(c) 2 V m
(d) π 2 V m

Q9. The average value of a sinusoidal waveform over a complete cycle is: [Topic: RMS value and average value of an AC signal | Easy]

(a) 0.637 V m
(b) Zero
(c) 0.707 V m
(d) V m

Q10. The average value of a sinusoidal waveform over a half cycle in terms of peak value V m is: [Topic: RMS value and average value of an AC signal | Easy]

(a) 0.637 V m
(b) 0.5 V m
(c) 0.318 V m
(d) 0.707 V m

Q11. The impedance of a circuit in complex (rectangular) form is written as: [Topic: complex representation of impedance | Easy]

(a) Z = R + j X
(b) Z = j R
(c) Z = R × X
(d) Z = R - R

Q12. In a purely inductive circuit, the inductive reactance is given by: [Topic: complex representation of impedance | Easy]

(a) X L = ω C 1
(b) X L = ω L 1
(c) X L = ω C
(d) X L = ω L

Q13. The capacitive reactance X C of a capacitor is expressed as: [Topic: complex representation of impedance | Easy]

(a) ω L
(b) ω L 1
(c) ω C 1
(d) ω C

Q14. In a series RL circuit, the current: [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Easy]

(a) Leads the applied voltage
(b) Is zero
(c) Lags the applied voltage
(d) Is in phase with the voltage

Q15. In a series RC circuit, the current: [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Easy]

(a) Lags the applied voltage
(b) Is always maximum
(c) Is in phase with the voltage
(d) Leads the applied voltage

Q16. The magnitude of impedance in a series RLC circuit is given by: [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Easy]

(a) R 2 + (X L - X C) 2
(b) R 2 + X L 2 + X C 2
(c) R 2 + (X L - X C) 2
(d) R + X L + X C

Q17. At resonance in a series RLC circuit, the net reactance is: [Topic: resonance in series RLC circuit | Easy]

(a) Infinite
(b) Equal to R
(c) Maximum
(d) Zero (X L = X C)

Q18. The resonant frequency of a series RLC circuit is given by: [Topic: resonance in series RLC circuit | Easy]

(a) f r = 2 π L C 1
(b) f r = 2 π L C 1
(c) f r = 2 π L C
(d) f r = 2 π L C

Q19. The power factor of an AC circuit is defined as: [Topic: power factor and power calculation in RL, RC and RLC circuits | Easy]

(a) sec ϕ
(b) tan ϕ
(c) sin ϕ
(d) cos ϕ

Q20. In a balanced three-phase star (Y) connection, the relation between line voltage and phase voltage is: [Topic: relations in line and phase voltages and currents in star and delta networks | Easy]

(a) V L = V p h
(b) V L = 3 V p h
(c) V L = 3 V p h
(d) V L = 3 V p h

Q21. A sinusoidal voltage has a peak value of 200 V . What is its RMS value? [Topic: RMS value and average value of an AC signal | Medium]

(a) 127.3 V
(b) 141.4 V
(c) 100 V
(d) 200 V

Q22. The average value of a full sinusoidal AC current over one complete cycle is: [Topic: RMS value and average value of an AC signal | Medium]

(a) Zero
(b) I m
(c) 0.707 I m
(d) 0.637 I m

Q23. The form factor of a pure sinusoidal waveform is approximately: [Topic: RMS value and average value of an AC signal | Medium]

(a) 0.637
(b) 0.707
(c) 1.11
(d) 1.414

Q24. In a pure capacitor, the current leads the applied voltage by a phase angle of: [Topic: phase difference | Medium]
(a) 4 5 °
(b) 9 0 °
(c) 0 °
(d) 18 0 °

Q25. A current is expressed as i = 10 sin (314 t + 3 0 °) A . What is the frequency of the supply? [Topic: phase | Medium]
(a) 100 Hz
(b) 60 Hz
(c) 50 Hz
(d) 314 Hz

Q26. A circuit has resistance R = 3 Ω and inductive reactance X L = 4 Ω . The magnitude of the impedance is: [Topic: complex representation of impedance | Medium]
(a) 5 Ω
(b) 7 Ω
(c) 1 Ω
(d) 25 Ω

Q27. The impedance of a series RC circuit is written in complex form as: [Topic: complex representation of impedance | Medium]
(a) Z = - R + j X C
(b) Z = R + j X C
(c) Z = R + j X L
(d) Z = R - j X C

Q28. In a series RL circuit, R = 10 Ω and X L = 10 Ω . The phase angle of the circuit is: [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Medium]
(a) 4 5 ° leading
(b) 6 0 ° lagging
(c) 4 5 ° lagging
(d) 3 0 ° lagging

Q29. A series RLC circuit has R = 6 Ω , X L = 12 Ω , and X C = 4 Ω . The net impedance magnitude is: [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Medium]
(a) 22 Ω
(b) 10 Ω
(c) 8 Ω
(d) 14 Ω

Q30. At resonance in a series RLC circuit, the impedance of the circuit is: [Topic: resonance in series RLC circuit | Medium]
(a) Minimum and purely reactive
(b) Maximum and purely reactive
(c) Minimum and purely resistive
(d) Maximum and purely resistive

Q31. The resonant frequency of a series RLC circuit with L = 2 H and C = 2 μ F is approximately: [Topic: resonance in series RLC circuit | Medium]
(a) 50 Hz
(b) 159 Hz

(c) 79.6 Hz
(d) 500 Hz

Q32. At resonance, the current in a series RLC circuit is in phase with the voltage because: [Topic: resonance in series RLC circuit | Medium]
(a) The current becomes zero
(b) The circuit behaves as purely resistive
(c) The circuit behaves as purely capacitive
(d) The circuit behaves as purely inductive

Q33. An RL circuit draws 5 A at 230 V with a power factor of 0.8 . The real power consumed is: [Topic: power factor and power calculation in RL, RC and RLC circuits | Medium]
(a) 1437 W
(b) 690 W
(c) 1150 W
(d) 920 W

Q34. The power factor of a circuit having impedance Z = 8 + j 6 Ω is: [Topic: power factor and power calculation in RL, RC and RLC circuits | Medium]
(a) 0.6 lagging
(b) 0.8 lagging
(c) 0.6 leading
(d) 0.8 leading

Q35. In a purely inductive AC circuit, the average power consumed over one cycle is: [Topic: power factor and power calculation in RL, RC and RLC circuits | Medium]
(a) 0.5 V I
(b) V I
(c) 0.707 V I
(d) Zero

Q36. In a balanced star-connected system, the relation between line voltage and phase voltage is: [Topic: relations in line and phase voltages and currents in star and delta networks | Medium]
(a) V L = V p h
(b) V L = 3 V p h
(c) V L = 3 V p h
(d) V L = 3 V p h

Q37. In a balanced delta-connected system, the relation between line current and phase current is: [Topic: relations in line and phase voltages and currents in star and delta networks | Medium]
(a) I L = 3 I p h
(b) I L = 3 I p h
(c) I L = I p h
(d) I L = 3 I p h

Q38. A balanced star-connected load has a line voltage of 400 V . The phase voltage is approximately: [Topic: relations in line and phase voltages and currents in star and delta networks | Medium]
(a) 692 V
(b) 133 V
(c) 231 V
(d) 400 V

Q39. In a three-phase balanced system, the phase difference between any two phase voltages is: [Topic: three-phase circuits- numbering and interconnection (delta or mesh connection) of three phases | Medium]
(a) 9 0 °
(b) 6 0 °

(c) 18 0 °
(d) 12 0 °

Q40. If an AC voltage completes 100 cycles in 2 seconds, its time period is: [Topic: alternating current and voltage | Medium]
(a) 0.5 s
(b) 2 s
(c) 50 s
(d) 0.02 s

Q41. A half-wave rectified sinusoidal current has a peak value of I m . What is the ratio of its RMS value to its average value? [Topic: RMS value and average value of an AC signal | Hard]
(a) 2 π
(b) π 2
(c) 2
(d) 2 π

Q42. A current waveform is given by i (t) = 10 + 5 sin (ω t) A. What is the RMS value of this current? [Topic: RMS value and average value of an AC signal | Hard]
(a) 112.5 A
(b) 10.6 A
(c) 125 A
(d) 12.5 A

Q43. A series RLC circuit has R = 10 Ω , L = 0.1 H, and C = 10 μ F. What is the quality factor Q at resonance? [Topic: resonance in series RLC circuit | Hard]
(a) 31.6
(b) 1
(c) 100
(d) 10

Q44. In a series RLC circuit at resonance, the voltage across the inductor is 50 V and across the resistor is 5 V. What is the quality factor Q ? [Topic: resonance in series RLC circuit | Hard]
(a) 5
(b) 10
(c) 50
(d) 0.1

Q45. An impedance is given as Z = 8 + j 6 Ω . If this is connected across a source V = 100∠3 0 ° V, what is the phase angle of the current? [Topic: complex representation of impedance | Hard]
(a) 36.8 7 °
(b) 66.8 7 °
(c) - 36.8 7 °
(d) - 6.8 7 °

Q46. A series RL circuit with R = 30 Ω and X L = 40 Ω is fed by 230 V, 50 Hz. What is the voltage across the inductor? [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Hard]
(a) 161 V
(b) 230 V
(c) 184 V
(d) 138 V

Q47. A single-phase load draws 10 A at 230 V with a power factor of 0.8 lagging. What is the reactive power consumed? [Topic: power factor and power calculation in RL, RC and RLC circuits | Hard]
(a) 920 VAR
(b) 1840 VAR
(c) 2300 VAR

(d) 1380 VAR

Q48. A balanced delta-connected load has a phase current of 10 A. If the same load impedances are reconnected in star across the same line voltage, what is the new line current? [Topic: relations in line and phase voltages and currents in star and delta networks | Hard]

(a) 3 10 3 A
(b) 30 A
(c) 10 3 A
(d) 10 A

Q49. In a balanced three-phase star-connected system, the line voltage is 400 V. What is the phase voltage? [Topic: relations in line and phase voltages and currents in star and delta networks | Hard]

(a) 400 V
(b) 693 V
(c) 566 V
(d) 231 V

Q50. A series RC circuit draws 5 A from a 100 V source and dissipates 300 W. What is the capacitive reactance X C ? [Topic: power factor and power calculation in RL, RC and RLC circuits | Hard]

(a) 20 Ω
(b) 8 Ω
(c) 16 Ω
(d) 12 Ω

Q51. A series RLC circuit has R = 5 Ω , X L = 20 Ω , X C = 12 Ω . For a supply of 100 V, what is the magnitude of the current and its nature? [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Hard]

(a) 5 A, leading
(b) 10.6 A, lagging
(c) 20 A, lagging
(d) 10.6 A, leading

Q52. A series RLC circuit resonates at 500 Hz with a bandwidth of 50 Hz. What are the approximate half-power frequencies? [Topic: resonance in series RLC circuit | Hard]

(a) 475 Hz and 525 Hz
(b) 495 Hz and 505 Hz
(c) 450 Hz and 550 Hz
(d) 400 Hz and 600 Hz

Q53. Two voltages are v 1 = 100 sin (ω t) and v 2 = 100 sin (ω t - π /3) . By what time (in terms of period T) does v 2 lag v 1 ? [Topic: phase difference | Hard]

(a) T /12
(b) T /3
(c) T /6
(d) T /2

Q54. Two impedances Z 1 = 10∠6 0 ° Ω and Z 2 = 10∠ - 3 0 ° Ω are connected in parallel. What is the magnitude of the equivalent impedance? [Topic: complex representation of impedance | Hard]

(a) 14.14 Ω
(b) 5 Ω
(c) 7.07 Ω
(d) 10 Ω

Q55. A coil draws 10 A when connected to a 120 V DC supply and 10 A when connected to a 230 V, 50 Hz AC supply. What is the power factor of the coil on AC? [Topic: power factor and power calculation in RL, RC and RLC circuits | Hard]

(a) 0.85
(b) 0.52

(c) 0.71
(d) 0.60

Q56. In a series RLC circuit, the applied voltage is 200 V, V R = 120 V and V L = 160 V. If V C = 0 , this is inconsistent; given V C = 320 V instead, what is V R consistent with 200 V supply? [Topic: steady state analysis of ac circuits consisting of RL, RC and RLC (series) | Hard]

(a) 160 V
(b) 200 V
(c) 120 V
(d) 80 V

Q57. A balanced delta-connected load of Z = 15∠3 0 ° Ω per phase is connected to a 400 V line. What is the line current? [Topic: relations in line and phase voltages and currents in star and delta networks | Hard]

(a) 80 A
(b) 46.2 A
(c) 26.7 A
(d) 15.4 A

Q58. A balanced three-phase star load draws a line current of 20 A at 400 V line voltage with power factor 0.866 . What is the total real power consumed? [Topic: power factor and power calculation in RL, RC and RLC circuits | Hard]

(a) 24 kW
(b) 8 kW
(c) 6.93 kW
(d) 12 kW

Q59. At resonance in a series RLC circuit, the current is 2 A and R = 20 Ω . If L = 50 mH and the resonant frequency is 1000 rad/s, what is the voltage across the capacitor? [Topic: resonance in series RLC circuit | Hard]

(a) 200 V
(b) 100 V
(c) 40 V
(d) 50 V

Q60. A sinusoidal voltage v (t) = 170 sin (377 t + 3 0 °) V is applied. What are the amplitude, frequency, and phase respectively? [Topic: definitions of amplitude, phase | Hard]

(a) 170 V, 377 Hz, 3 0 °
(b) 170 V, 60 Hz, 3 0 °
(c) 120 V, 60 Hz, 3 0 °
(d) 170 V, 50 Hz, 0.52 rad

2.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 2 **[Subjective & Numerical Evaluation]**

1 A series RLC circuit consists of resistance $R = 10\,\Omega$, inductance $L = 50$ mH, and capacitance $C = 20\,\mu$F connected across a sinusoidal voltage supply $v(t) = 230\sqrt{2}\sin(100\pi t)$ V. Determine:

(a) The resonant frequency f_0, quality factor Q_0, and bandwidth BW of the circuit.

(b) The circuit impedance $\mathbf{Z}$, current $\mathbf{I}$, power factor $\cos\phi$, and real power P absorbed at supply frequency $f = 50$ Hz.

(c) The half-power frequencies f_1 and f_2.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Resonant Frequency, Quality Factor, and Bandwidth
The resonant angular frequency is:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 20 \times 10^{-6}}} = 1000 \text{ rad/s}$$

The resonant cyclic frequency is $f_0 = \frac{\omega_0}{2\pi} = \frac{1000}{2\pi} \approx 159.15$ Hz.The Quality Factor Q_0 is given by:

$$Q_0 = \frac{\omega_0 L}{R} = \frac{1000 \times 0.050}{10} = 5.0$$

The Bandwidth in Hertz is:

$$BW = \frac{f_0}{Q_0} = \frac{159.15}{5.0} \approx 31.83 \text{ Hz}$$

Step 2: Circuit Behavior at $f = 50$ Hz ($\omega = 100\pi \approx 314.16$ rad/s)
Compute inductive and capacitive reactances:

$$X_L = \omega L = 314.16 \times 0.050 = 15.71\,\Omega$$
$$X_C = \frac{1}{\omega C} = \frac{1}{314.16 \times 20 \times 10^{-6}} \approx 159.15\,\Omega$$

Net reactance is $X = X_L - X_C = 15.71 - 159.15 = -143.44\,\Omega$ (capacitive).Complex impedance: $\mathbf{Z} = 10 - j143.44\,\Omega = 143.79\angle -86.0^\circ\,\Omega$.RMS current magnitude: $I = \frac{V_{rms}}{|\mathbf{Z}|} = \frac{230}{143.79} \approx 1.60$ A.Power factor: $\cos\phi = \frac{R}{|\mathbf{Z}|} = \frac{10}{143.79} \approx 0.0695$ leading.Real power: $P = VI\cos\phi = 230 \times 1.60 \times 0.0695 \approx 25.6$ W ($P = I^2R = 1.60^2 \times 10 = 25.6$ W).

Step 3: Half-Power Frequencies

$$f_1 = f_0 - \frac{BW}{2} = 159.15 - 15.92 = 143.23 \text{ Hz}, \quad f_2 = f_0 + \frac{BW}{2} = 159.15 + 15.92 = 175.07 \text{ Hz}$$

Key Examiner Caveat: At frequencies below resonance ($\omega < \omega_0$), $X_C > X_L$, making the circuit predominantly capacitive and current leading.

Chapter 3

Unit 3: Fundamentals of Electrical Machines

3.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. Mutual inductance between two coils is the property by which a changing current in one coil induces an EMF in the: [Topic: mutual inductance and mutual coupling phenomena in transformer | Easy]

(a) Same coil only
(b) Iron core only
(c) Neighbouring coil
(d) Supply source

Q2. The SI unit of mutual inductance is the: [Topic: mutual inductance and mutual coupling phenomena in transformer | Easy]

(a) Farad
(b) Henry
(c) Weber
(d) Tesla

Q3. In a transformer, magnetic coupling between the two windings is provided mainly by the: [Topic: mutual inductance and mutual coupling phenomena in transformer | Easy]

(a) Laminated iron core
(b) Insulation layer
(c) Copper losses
(d) Air gap

Q4. A transformer works on the principle of: [Topic: transformer – working | Easy]

(a) Mutual electromagnetic induction
(b) Self capacitance
(c) Chemical reaction
(d) Electrostatic induction

Q5. A transformer can operate only with: [Topic: transformer – working | Easy]

(a) Direct current
(b) Static charge
(c) Alternating current
(d) Pulsating constant current

Q6. A transformer that increases the output voltage is called a: [Topic: transformer – working | Easy]

(a) Current transformer
(b) Isolation transformer
(c) Step-down transformer

(d) Step-up transformer

Q7. The turns ratio of a transformer is defined as the ratio of: [Topic: concept of turns ratio and applications | Easy]

(a) Primary power to secondary power
(b) Primary turns to secondary turns
(c) Core area to window area
(d) Primary current to secondary current

Q8. For an ideal transformer, the voltage ratio equals the turns ratio as V 2 V 1 = ? [Topic: concept of turns ratio and applications | Easy]

(a) N 1 N 2 1
(b) N 2 N 1
(c) N 1 × N 2
(d) N 1 N 2

Q9. In an ideal transformer, if the voltage is stepped up, the current in that winding is: [Topic: concept of turns ratio and applications | Easy]

(a) Reduced to zero
(b) Unchanged
(c) Stepped up equally
(d) Stepped down

Q10. The device used to convert the alternating EMF in a DC generator armature into DC at the output is the: [Topic: dc machines- working principles, classification | Easy]

(a) Transformer
(b) Commutator
(c) Slip ring
(d) Capacitor

Q11. Based on the field connection, DC machines are commonly classified as series, shunt and: [Topic: dc machines- working principles, classification | Easy]

(a) Compound
(b) Synchronous
(c) Parallel
(d) Induction

Q12. A DC motor converts electrical energy into: [Topic: dc machines- working principles, classification | Easy]

(a) Light energy
(b) Chemical energy
(c) Mechanical energy
(d) Heat energy only

Q13. A starter is used with a DC motor mainly to limit the: [Topic: starting, speed control and applications of dc motors | Easy]

(a) Field flux
(b) Running speed
(c) Output torque
(d) High starting current

Q14. Which DC motor is best suited for applications requiring very high starting torque, such as traction? [Topic: starting, speed control and applications of dc motors | Easy]

(a) Series motor
(b) Shunt motor
(c) Synchronous motor
(d) Separately excited motor

Q15. The rotating magnetic field in a three-phase induction motor is produced by the: [Topic: working principle of single and three phase induction motors | Easy]

(a) Three-phase stator windings
(b) Permanent magnets
(c) Commutator
(d) DC field supply

Q16. A single-phase induction motor is not self-starting because its stator produces a: [Topic: working principle of single and three phase induction motors | Easy]

(a) Rotating magnetic field
(b) Zero magnetic field
(c) Constant DC field
(d) Pulsating magnetic field

Q17. Three-phase induction motors are widely used in industry mainly because they are: [Topic: applications of ac motors | Easy]

(a) Expensive and complex
(b) Rugged and low maintenance
(c) Suitable for DC supply
(d) Unable to run on load

Q18. Fleming's left-hand rule is used to find the direction of force in a: [Topic: Fleming's left hand and right hand rule | Easy]

(a) Transformer
(b) Generator
(c) Rectifier
(d) Motor

Q19. In Fleming's right-hand rule, the thumb represents the direction of: [Topic: Fleming's left hand and right hand rule | Easy]

(a) Motion of the conductor
(b) Magnetic field
(c) Applied force
(d) Induced current

Q20. If a DC voltage is applied to a transformer primary, the output voltage at the secondary will be: [Topic: transformer on DC | Easy]

(a) Zero in steady state
(b) Negative of input
(c) Doubled
(d) Equal to input

Q21. A single-phase transformer has 400 primary turns and 100 secondary turns. If the primary voltage is 230 V , what is the secondary voltage (assuming ideal transformer)? [Topic: concept of turns ratio and applications | Medium]

(a) 460 V
(b) 57.5 V
(c) 115 V
(d) 920 V

Q22. An ideal step-up transformer increases voltage by a factor of 5. If the primary current is 10 A , the secondary current is: [Topic: concept of turns ratio and applications | Medium]

(a) 50 A
(b) 10 A
(c) 2 A
(d) 5 A

Q23. What happens when a DC supply is connected to the primary of a transformer? [Topic:

transformer on DC | Medium]

(a) The secondary current reverses direction periodically
(b) The transformer works normally with slightly reduced efficiency
(c) A steady voltage appears across the secondary equal to the turns ratio times the input
(d) The constant flux induces no EMF in the secondary, and the low winding resistance causes a very large steady current that can overheat and burn the primary coil

Q24. Two coils have a mutual inductance of 0.2 H . If the current in one coil changes at a rate of 500 A/s , the EMF induced in the other coil is: [Topic: mutual inductance and mutual coupling phenomena in transformer | Medium]

(a) 10 V
(b) 2500 V
(c) 250 V
(d) 100 V

Q25. Two coils with self-inductances L 1 = 0.1 H and L 2 = 0.4 H are perfectly coupled (k = 1). The mutual inductance is: [Topic: mutual inductance and mutual coupling phenomena in transformer | Medium]

(a) 0.2 H
(b) 0.1 H
(c) 0.25 H
(d) 0.5 H

Q26. The primary function of the laminated iron core in a transformer is to: [Topic: transformer – working | Medium]

(a) Convert AC into pulsating DC
(b) Provide a low-reluctance path for the magnetic flux while reducing eddy current losses
(c) Store electrical charge between the windings
(d) Increase the resistance of the windings

Q27. The EMF equation of a transformer is E = 4.44 f N ϕ m . If the supply frequency is doubled while keeping voltage and turns constant, the maximum flux ϕ m will: [Topic: transformer – working | Medium]

(a) Reduce to half
(b) Double
(c) Reduce to one-fourth
(d) Remain the same

Q28. Compared to a two-winding transformer of the same rating, an auto-transformer requires: [Topic: auto-transformer | Medium]

(a) The same amount of copper but larger core
(b) Less copper and has higher efficiency, but offers no electrical isolation between primary and secondary
(c) More copper and has lower efficiency
(d) Two completely separate windings with full isolation

Q29. A current transformer (CT) is used to measure large currents. During operation, its secondary should: [Topic: instrument transformers | Medium]

(a) Never be left open-circuited, as dangerously high voltages can be induced
(b) Be connected to the supply voltage directly
(c) Be connected across a high-value resistor only
(d) Always be left open-circuited for safety

Q30. A potential transformer (PT) with ratio 11000/110 V shows 100 V on its voltmeter. The actual line voltage is: [Topic: instrument transformers | Medium]

(a) 1100 V
(b) 1000 V
(c) 10000 V
(d) 11000 V

Q31. Fleming's left-hand rule is used to determine the direction of: [Topic: Fleming's left hand and right hand rule | Medium]

(a) Force on a current-carrying conductor in a magnetic field (motor action)
(b) Induced EMF in a generator
(c) Magnetic flux inside a solenoid
(d) The resistance of a conductor

Q32. In Fleming's right-hand rule, the forefinger, thumb, and middle finger respectively represent: [Topic: Fleming's left hand and right hand rule | Medium]

(a) Current, field, and motion
(b) Motion, current, and field
(c) Field, motion, and induced current
(d) Field, current, and force

Q33. In a DC motor, the back EMF (E b) at the instant of starting is: [Topic: dc machines- working principles, classification, starting, speed control and applications of dc motors | Medium]

(a) Maximum, equal to the supply voltage
(b) Zero, because the armature is not yet rotating
(c) Negative and opposing the current
(d) Equal to half the supply voltage

Q34. A DC shunt motor runs at 1000 rpm . If the flux is reduced to 80% of its original value (armature EMF roughly constant), the speed becomes approximately: [Topic: dc machines- working principles, classification, starting, speed control and applications of dc motors | Medium]

(a) 1250 rpm
(b) 640 rpm
(c) 800 rpm
(d) 1000 rpm

Q35. Which DC motor is best suited for applications requiring high starting torque, such as electric traction and cranes? [Topic: dc machines- working principles, classification, starting, speed control and applications of dc motors | Medium]

(a) DC series motor
(b) DC compound (cumulative) motor
(c) Separately excited DC motor
(d) DC shunt motor

Q36. A DC series motor should never be started without a mechanical load because: [Topic: dc machines- working principles, classification, starting, speed control and applications of dc motors | Medium]

(a) The armature current becomes zero
(b) At no load the flux is very low, causing the speed to rise dangerously high and risk mechanical damage
(c) The commutator will stop working
(d) It will fail to develop any torque at all

Q37. A 4-pole, three-phase induction motor is connected to a 50 Hz supply. Its synchronous speed is: [Topic: working principle of single and three phase induction motors | Medium]

(a) 1000 rpm
(b) 750 rpm
(c) 1500 rpm
(d) 3000 rpm

Q38. A three-phase induction motor has a synchronous speed of 1500 rpm and runs at 1440 rpm . The percentage slip is: [Topic: working principle of single and three phase induction motors | Medium]

(a) 10%
(b) 4%
(c) 6%
(d) 2%

Q39. A single-phase induction motor is not self-starting because: [Topic: working principle of single and three phase induction motors | Medium]

(a) The supply frequency is too low to start it
(b) It has no rotor conductors
(c) The rotor is made of a permanent magnet
(d) The single-phase winding produces a pulsating field, giving zero net starting torque

Q40. Which AC motor is most commonly used in fans, pumps, and household appliances due to its ruggedness, low cost, and low maintenance? [Topic: applications of ac motors | Medium]

(a) Synchronous motor
(b) Wound-rotor induction motor
(c) Universal motor
(d) Squirrel-cage induction motor

Q41. Two coils have self-inductances L 1 = 0.4 H and L 2 = 0.9 H . When connected in series aiding, the total inductance is 1.78 H . What is the coefficient of coupling k ? [Topic: mutual inductance and mutual coupling phenomena in transformer | Hard]

(a) 0.80
(b) 0.65
(c) 0.40
(d) 0.50

Q42. A transformer rated for 230 V AC is accidentally connected to a 230 V DC supply. Which outcome best describes what happens and why? [Topic: transformer on DC | Hard]

(a) The primary draws a very large current limited only by its small winding resistance, risking burnout, because DC produces no back-EMF of self-induction in steady state and the inductive reactance is zero
(b) The core saturates but current stays at rated value
(c) The secondary delivers 230 V DC safely
(d) The transformer steps the DC voltage up or down normally

Q43. A step-down transformer has turns ratio 10 : 1 . The secondary is loaded with 5 Ω . What impedance is reflected to the primary? [Topic: concept of turns ratio and applications | Hard]

(a) 250 Ω
(b) 0.05 Ω
(c) 500 Ω
(d) 50 Ω

Q44. In a practical transformer at no-load, the no-load current lags the applied voltage by nearly 8 0 ° . What does this large angle primarily indicate? [Topic: transformer – working | Hard]

(a) The core losses are extremely high
(b) The magnetizing (reactive) component dominates over the small core-loss (active) component
(c) The transformer is heavily loaded
(d) The winding resistance is dominant

Q45. A 1 kV A , 200/100 V step-down transformer is reconnected as an auto-transformer to give 200/300 V . What is the new kVA rating as an auto-transformer? [Topic: auto-transformer | Hard]

(a) 1 kV A
(b) 2 kV A
(c) 1.5 kV A
(d) 3 kV A

Q46. Compared to an equivalent two-winding transformer of the same output, an auto-transformer transfers part of the power conductively. For a turns ratio where N 1 N 2 = 0.8 (step-down), what fraction of the total power is transferred inductively (transformed)? [Topic: auto-transformer | Hard]

(a) 1.0
(b) 0.2
(c) 0.5
(d) 0.8

Q47. Why should the secondary of a current transformer (CT) never be left open-circuited while the primary carries current? [Topic: instrument transformers | Hard]

(a) The absence of secondary demagnetizing MMT lets the full primary MMT magnetize the core, producing dangerously high flux, core overheating and a very high secondary voltage
(b) The secondary voltage drops to zero
(c) The CT ratio simply becomes inaccurate but stays safe
(d) The primary current falls to zero immediately

Q48. A potential transformer (PT) is used to measure a 11 kV line voltage with a standard 110 V voltmeter. If the measured ratio error causes the voltmeter to read 108 V for an actual 11 kV , what is the approximate percentage ratio error? [Topic: instrument transformers | Hard]

(a) + 2.0%
(b) + 1.8%
(c) - 2.0%
(d) - 1.8%

Q49. In a DC generator, the function of the commutator is best described as: [Topic: dc machines-working principles | Hard]

(a) A mechanical rectifier that converts the internally generated alternating EMF into unidirectional (DC) output at the brushes
(b) A device that increases the generated EMF
(c) A field-weakening control element
(d) A frequency converter for the AC output

Q50. A DC shunt motor runs at 1000 r p m drawing an armature current of 20 A from a 220 V supply with armature resistance 0.5 Ω . If flux is held constant and the load torque is unchanged, what series armature resistance must be added to reduce the speed to 800 r p m ? [Topic: speed control and applications of dc motors | Hard]

(a) 2.1 Ω
(b) 3.0 Ω
(c) 1.5 Ω
(d) 0.5 Ω

Q51. A 250 V DC shunt motor has armature resistance 0.4 Ω . To limit the starting current to twice the full-load current of 40 A , what starter resistance is required at the instant of starting? [Topic: starting | Hard]

(a) 2.5 Ω
(b) 2.725 Ω
(c) 6.25 Ω
(d) 3.125 Ω

Q52. In a DC series motor, if the load torque suddenly drops to nearly zero (no-load), what happens and why? [Topic: classification | Hard]

(a) The speed drops to zero
(b) The speed stays constant due to flux regulation
(c) The motor stalls immediately
(d) The speed rises dangerously high (racing) because the flux falls with the small current, and since $N \propto 1/\phi$ at fixed voltage the speed increases without bound

Q53. A 4 -pole, 50 Hz three-phase induction motor runs at 1440 r p m . What is the frequency of the rotor-induced EMF? [Topic: working principle of single and three phase induction motors | Hard]

(a) 4 Hz
(b) 2 Hz
(c) 48 Hz
(d) 50 Hz

Q54. Why is a single-phase induction motor not self-starting, according to the double-field revolving theory? [Topic: working principle of single and three phase induction motors | Hard]

(a) The pulsating field splits into two equal and opposite rotating fields that produce equal and opposite torques, giving zero net starting torque at standstill
(b) The rotor bars are open-circuited at start
(c) The stator produces no magnetic field at start
(d) The supply frequency is too low to induce rotor current

Q55. A conductor carries current in the + x direction and lies in a magnetic field pointing in the + y direction. Using Fleming's left-hand rule, in which direction is the force on the conductor? [Topic: Fleming's left hand and right hand rule | Hard]

(a) + z direction
(b) - x direction
(c) - z direction
(d) + y direction

Q56. Which statement correctly distinguishes the two Fleming's rules in the context of electrical machines? [Topic: Fleming's left hand and right hand rule | Hard]

(a) Both rules give force direction, differing only by hand
(b) Both rules apply only to transformers, not rotating machines
(c) The right-hand rule gives induced EMF/current direction in a generator; the left-hand rule gives force direction in a motor
(d) The left-hand rule applies to generators; the right-hand rule applies to motors

Q57. For a constant-speed drive such as a large industrial fan or pump where power factor correction is also desired, which AC motor is most suitable? [Topic: applications of ac motors | Hard]

(a) Capacitor-start induction motor
(b) Universal motor
(c) Synchronous motor, since it runs at constant speed and can be over-excited to deliver leading reactive power
(d) Single-phase shaded-pole motor

Q58. A transformer has iron loss 400 W and full-load copper loss 400 W . At what fraction of full load does maximum efficiency occur, and how does the copper loss there compare with the iron loss? [Topic: transformer – working | Hard]

(a) At full load; copper loss is double the iron loss
(b) At half load; copper loss equals 200 W
(c) At quarter load; copper loss equals iron loss
(d) At full load; copper loss equals iron loss (400 W each)

Q59. In field (flux) control of a DC shunt motor above base speed, the output power is approximately constant. If flux is reduced to 0.5 times its original value with armature voltage and current held at rated, the torque capability becomes: [Topic: speed control and applications of dc motors | Hard]

(a) Halved, since torque $\propto \phi$ I a while speed roughly doubles keeping power constant
(b) Reduced to one quarter
(c) Unchanged
(d) Doubled

Q60. Two magnetically coupled coils connected in series carry the same current. When connected series-aiding the total inductance is 2.2 H and when series-opposing it is 1.0 H . What is the mutual inductance M ? [Topic: mutual inductance and mutual coupling phenomena in transformer | Hard]

(a) 0.15 H
(b) 1.2 H
(c) 0.3 H
(d) 0.6 H

3.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 3 **[Subjective & Numerical Evaluation]**

1 A 25 kVA, 2200/220 V, 50 Hz single-phase distribution transformer has iron loss of 350 W and full-load copper loss of 400 W. Calculate:

(a) The efficiency at full-load and 0.8 power factor lagging.

(b) The load fraction x at which maximum efficiency occurs, and the value of maximum efficiency at unity power factor.

(c) Explain why transformers cannot operate on a steady D.C. supply.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Efficiency at Full Load ($x = 1.0, \cos\phi = 0.8$)
Output power: $P_{out} = S \times \cos\phi = 25000 \times 0.8 = 20000$ W $= 20$ kW.Total losses: $P_{losses} = P_i + P_{cu} = 350$ W $+ 400$ W $= 750$ W $= 0.75$ kW.

$$\eta_{FL} = \frac{20000}{20000 + 750} \times 100\% \approx 96.39\%$$

Step 2: Maximum Efficiency Condition
Maximum efficiency occurs when copper loss equals iron loss ($x^2 P_{cu} = P_i$):

$$x = \sqrt{\frac{P_i}{P_{cu}}} = \sqrt{\frac{350}{400}} \approx 0.9354 \text{ (at } 93.54\% \text{ of rated load)}$$

At this fraction and unity power factor ($\cos\phi = 1.0$):

$$P_{out} = 0.9354 \times 25000 \times 1.0 = 23385 \text{ W}$$

Total losses: $P_{losses} = 2P_i = 700$ W.

$$\eta_{max} = \frac{23385}{23385 + 700} \times 100\% \approx 97.09\%$$

Step 3: Phenomenon of Transformer on Steady D.C.
Faraday's Law states induced EMF is $e = -N\frac{d\Phi}{dt}$. With steady D.C., $\frac{d\Phi}{dt} = 0$, so no counter-EMF is generated in the primary winding. The primary current is limited solely by low winding resistance R_1: $I_{dc} = V_{dc}/R_1$. This draws massive inrush current, causing immediate core saturation, extreme Joule heating, insulation breakdown, and transformer destruction.

Marking Allocation: Full load calculation: **3.5 Marks**; Condition for maximum efficiency: **3.5 Marks**; DC saturation explanation: **3.0 Marks**.

Chapter 4

Unit 4: Fundamentals of Semiconductor Devices

4.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. Which of the following is the most commonly used semiconductor material in electronic devices? [Topic: introduction to semiconductor devices | Easy]

(a) Aluminium
(b) Silicon
(c) Copper
(d) Mica

Q2. A pure semiconductor without any impurity added is known as a: [Topic: introduction to semiconductor devices | Easy]

(a) P-type semiconductor
(b) Extrinsic semiconductor
(c) N-type semiconductor
(d) Intrinsic semiconductor

Q3. In an N-type semiconductor, the majority charge carriers are: [Topic: introduction to semiconductor devices | Easy]

(a) Protons
(b) Neutrons
(c) Electrons
(d) Holes

Q4. When a PN junction diode is forward biased, the depletion region: [Topic: Working and operation of PN Junction diode | Easy]

(a) Decreases in width
(b) Increases in width
(c) Becomes infinitely wide
(d) Stays exactly the same

Q5. A PN junction diode primarily allows current to flow in: [Topic: Working and operation of PN Junction diode | Easy]

(a) No direction at all
(b) Both directions equally
(c) Only one direction
(d) Only alternating directions

Q6. The approximate forward voltage drop (knee voltage) of a silicon PN junction diode is about: [Topic: Characteristics of PN junction diode | Easy]

(a) 1.5 V
(b) 5.0 V
(c) 0.3 V
(d) 0.7 V

Q7. In the reverse-biased region of an ideal PN junction diode, the current is: [Topic: Characteristics of PN junction diode | Easy]
(a) Equal to forward current
(b) Nearly zero
(c) Very large
(d) Infinite

Q8. A Zener diode is normally operated in which region? [Topic: working and operation of Zener diode | Easy]
(a) Reverse breakdown region
(b) Cut-off region only
(c) Saturation region
(d) Forward conduction region

Q9. The main application of a Zener diode is as a: [Topic: working and operation of Zener diode | Easy]
(a) Current amplifier
(b) Signal oscillator
(c) Voltage regulator
(d) Light emitter

Q10. In the Zener breakdown region, the voltage across the Zener diode remains: [Topic: characteristics of Zener diode | Easy]
(a) Rapidly increasing
(b) Steadily decreasing
(c) Equal to zero
(d) Almost constant

Q11. Which logic gate produces an output of 1 only when all its inputs are 1 ? [Topic: logic gates | Easy]
(a) NOR gate
(b) NOT gate
(c) OR gate
(d) AND gate

Q12. A NOT gate is also commonly called a(n): [Topic: logic gates | Easy]
(a) Adder
(b) Inverter
(c) Comparator
(d) Buffer

Q13. The output of a two-input OR gate is 0 only when: [Topic: logic gates | Easy]
(a) Both inputs are 0
(b) One input is 1
(c) Any input is 1
(d) Both inputs are 1

Q14. A varactor diode is used to provide a voltage-controlled: [Topic: Working and Operation of Varactor diode | Easy]
(a) Inductance
(b) Resistance
(c) Capacitance
(d) Frequency source

Q15. A varactor diode is normally operated under: [Topic: Working and Operation of Varactor diode | Easy]

(a) Forward bias
(b) Zero bias only
(c) No bias at all
(d) Reverse bias

Q16. When testing a good diode with a multimeter, it shows low resistance in forward direction and high resistance in reverse. A diode that reads low resistance in both directions is: [Topic: Testing of diode and BJT | Easy]

(a) Reverse biased
(b) Shorted
(c) Perfectly good
(d) Open

Q17. A half-wave rectifier conducts during: [Topic: Half wave and Full wave rectifier | Easy]

(a) Only one half of the AC cycle
(b) Only the negative peaks
(c) Both halves of the AC cycle
(d) The full cycle continuously

Q18. Compared to a half-wave rectifier, a full-wave rectifier has: [Topic: Half wave and Full wave rectifier | Easy]

(a) Higher output efficiency
(b) The same single diode
(c) A
(d) No output voltage

Q19. How many diodes are used in a full-wave bridge rectifier? [Topic: Full wave bridge rectifier | Easy]

(a) Two
(b) Four
(c) One
(d) Six

Q20. What does the abbreviation MOSFET stand for? [Topic: MOSFET representation and characteristics | Easy]

(a) Metal Only Semiconductor Fast Emitting Transistor
(b) Multiple Output Switching Field Effect Transistor
(c) Metal Oxide Semiconductor Field Effect Transistor
(d) Metal Oxide Silicon Forward Emitter Transistor

Q21. A pure silicon crystal is doped with a pentavalent impurity such as phosphorus. What type of semiconductor is formed and what are the majority carriers? [Topic: introduction to semiconductor devices | Medium]

(a) Intrinsic semiconductor with equal electrons and holes
(b) N-type with electrons as majority carriers
(c) P-type with holes as majority carriers
(d) N-type with holes as majority carriers

Q22. When a PN junction diode is forward biased, what happens to the width of the depletion region and the potential barrier? [Topic: Working and operation of PN Junction diode | Medium]

(a) Both increase, blocking current flow
(b) Width increases but barrier decreases
(c) Both decrease, allowing current to flow
(d) Width decreases but barrier increases

Q23. In the V-I characteristics of a silicon PN junction diode, appreciable forward current begins

only after the applied voltage exceeds approximately which value? [Topic: Characteristics of PN junction diode | Medium]

(a) 1.1 V
(b) 0.0 V
(c) 0.7 V
(d) 0.3 V

Q24. A Zener diode used for voltage regulation is always connected in which configuration in the circuit? [Topic: working and operation of Zener diode | Medium]

(a) Reverse biased and operated in the breakdown region
(b) Reverse biased but operated below breakdown
(c) Forward biased near its knee voltage
(d) Forward biased and operated in saturation

Q25. In the reverse breakdown region of a Zener diode's characteristic curve, how do voltage and current behave? [Topic: characteristics of Zener diode | Medium]

(a) Voltage increases rapidly while current stays constant
(b) Both voltage and current decrease together
(c) Voltage remains almost constant while current changes significantly
(d) Both voltage and current increase proportionally

Q26. A two-input logic gate produces a HIGH output only when both inputs are LOW. Which gate is this? [Topic: logic gates | Medium]

(a) XOR gate
(b) AND gate
(c) NOR gate
(d) NAND gate

Q27. Using De Morgan's theorem, the expression $A \cdot B$ is equivalent to which of the following? [Topic: logic gates | Medium]

(a) $A + B$
(b) $A \cdot B$
(c) $A \cdot B$
(d) $A + B$

Q28. A varactor diode is operated under reverse bias so that it behaves like a voltage-controlled capacitor. As the reverse voltage increases, what happens to its capacitance? [Topic: Working and Operation of Varactor diode | Medium]

(a) Capacitance increases because the depletion width increases
(b) Capacitance decreases because the depletion width increases
(c) Capacitance stays constant regardless of reverse voltage
(d) Capacitance increases because the depletion width decreases

Q29. While testing a diode with a digital multimeter in diode mode, a technician reads about 0.7 V in one direction and "OL" (open) when the leads are reversed. What is the condition of the diode? [Topic: Testing of diode and BJT | Medium]

(a) The diode is leaky
(b) The diode is shorted
(c) The diode is open in both directions
(d) The diode is good

Q30. When testing an NPN transistor with a multimeter, the base-emitter and base-collector junctions should behave like which arrangement of diodes? [Topic: Testing of diode and BJT | Medium]

(a) Two diodes with their anodes joined at the base
(b) Two diodes with their cathodes joined at the base
(c) Two diodes connected in series between collector and emitter
(d) A single diode between collector and emitter

Q31. A half-wave rectifier is supplied with a 50 Hz AC input. What is the ripple frequency of its output? [Topic: Half wave and Full wave rectifier | Medium]

(a) 100 Hz
(b) 150 Hz
(c) 50 Hz
(d) 25 Hz

Q32. A full-wave rectifier produces a peak output voltage of 10 V . What is the approximate average (DC) output voltage? [Topic: Half wave and Full wave rectifier | Medium]

(a) 10 V
(b) 6.37 V
(c) 7.07 V
(d) 3.18 V

Q33. What is the maximum theoretical rectification efficiency of a full-wave rectifier? [Topic: Half wave and Full wave rectifier | Medium]

(a) 50%
(b) 40.6%
(c) 81.2%
(d) 100%

Q34. In a full-wave bridge rectifier using a single transformer secondary of peak voltage V m , approximately what is the Peak Inverse Voltage (PIV) across each diode? [Topic: Full wave bridge rectifier | Medium]

(a) 0.5 V m
(b) 2 V m
(c) V m
(d) V m /2

Q35. A key advantage of a bridge rectifier over a center-tapped full-wave rectifier is that it: [Topic: Full wave bridge rectifier | Medium]

(a) Produces a higher ripple frequency
(b) Requires no transformer at all
(c) Does not require a center-tapped transformer
(d) Uses only one diode

Q36. An enhancement-type N-channel MOSFET starts conducting between drain and source only when the gate-to-source voltage satisfies which condition? [Topic: MOSFET representation and characteristics | Medium]

(a) V GS $<$ 0
(b) V GS = V D S
(c) V GS $>$ V t h (threshold voltage)
(d) V GS = 0

Q37. In a MOSFET, the gate terminal controls the drain current but draws almost no DC current because: [Topic: MOSFET representation and characteristics | Medium]

(a) The gate forms a forward-biased PN junction
(b) The gate is insulated from the channel by a thin oxide layer
(c) The gate is made of a highly resistive metal
(d) The gate is directly connected to the source

Q38. In a power supply circuit, a capacitor is connected across the output of a rectifier. Its primary function is to: [Topic: Application of diodes, transistors, rectifiers and MOSFET | Medium]

(a) Convert AC into DC
(b) Rectify the negative half-cycles
(c) Smooth the output by reducing ripple
(d) Increase the peak output voltage indefinitely

Q39. A diode is placed across the coil of a relay in a circuit driven by a transistor. The purpose of this diode is to: [Topic: Application of diodes, transistors, rectifiers and MOSFET | Medium]

(a) Act as a voltage regulator for the coil
(b) Protect the transistor by suppressing the inductive voltage spike when the coil switches off
(c) Increase the current flowing through the relay coil
(d) Rectify the AC supply feeding the relay

Q40. The reverse saturation current of a PN junction diode is primarily due to: [Topic: Characteristics of PN junction diode | Medium]

(a) The applied forward voltage
(b) Minority carriers crossing the junction
(c) The barrier potential increasing
(d) Majority carriers crossing the junction

Q41. A silicon diode has a reverse saturation current I S = 10 nA at 300 K . Using the ideal diode equation with V T = 25.9 mV and ideality factor n = 1 , what forward voltage is required to produce a current of 1 mA ? [Topic: Characteristics of PN junction diode | Hard]

(a) Approximately 0.477 V
(b) Approximately 0.600 V
(c) Approximately 0.298 V
(d) Approximately 0.700 V

Q42. In a PN junction at equilibrium, the built-in potential barrier opposes diffusion of majority carriers. If the doping on both sides is increased by a factor of 10, how does the built-in potential V bi change (temperature constant)? [Topic: Working and operation of PN Junction diode | Hard]

(a) It increases by V T ln 10
(b) It increases by 2 V T ln 10
(c) It remains unchanged
(d) It decreases by 2 V T ln 10

Q43. A Zener regulator uses a 6.2 V Zener with dynamic resistance r Z = 5 Ω . If the Zener current changes from 5 mA to 25 mA , what is the change in output (Zener) voltage? [Topic: characteristics of Zener diode | Hard]

(a) 0.10 V
(b) 0.05 V
(c) 0.125 V
(d) 0.25 V

Q44. In a shunt Zener regulator with V in = 15 V , series resistor R S = 220 Ω , and a 9 V Zener, what is the maximum load current the circuit can supply while keeping the Zener in regulation (ignore minimum Zener current)? [Topic: working and operation of Zener diode | Hard]

(a) Approximately 27.3 mA
(b) Approximately 68.2 mA
(c) Approximately 40.9 mA
(d) Approximately 18.0 mA

Q45. A half-wave rectifier delivers a peak output of 20 V into a resistive load. What is the ripple factor of the output? [Topic: Half wave and Full wave rectifier | Hard]

(a) 0.318
(b) 0.636
(c) 0.482
(d) 1.21

Q46. A full-wave bridge rectifier is fed from a transformer secondary with RMS voltage 18 V . Assuming silicon diodes with 0.7 V drop each, what is the approximate DC output voltage (no filter)? [Topic: Full wave bridge rectifier | Hard]

(a) Approximately 16.2 V
(b) Approximately 7.6 V

(c) Approximately 11.5 V
(d) Approximately 15.3 V

Q47. For a full-wave bridge rectifier, how does the Peak Inverse Voltage (PIV) rating per diode compare to a center-tapped full-wave rectifier producing the same peak output V m ? [Topic: Full wave bridge rectifier | Hard]

(a) Bridge PIV = V m , center-tapped PIV = 2 V m
(b) Both have PIV = V m
(c) Both have PIV = 2 V m
(d) Bridge PIV = 2 V m , center-tapped PIV = V m

Q48. A varactor diode has junction capacitance C j = C 0 / (1 + V R / V bi) 1/2 with C 0 = 80 pF and V bi = 0.7 V . What is C j at a reverse bias of V R = 8.3 V ? [Topic: Working and Operation of Varactor diode | Hard]

(a) 40.0 pF
(b) 24.6 pF
(c) 56.6 pF
(d) 8.0 pF

Q49. A logic function is given by F = A + B + A · B . Which single gate is equivalent to F ? [Topic: logic gates | Hard]

(a) A NOR gate
(b) An XOR gate
(c) An AND gate
(d) A NOT gate on A (i.e., F = A)

Q50. Using only 2-input NAND gates, what is the minimum number of NAND gates required to implement a 2-input XOR function? [Topic: logic gates | Hard]

(a) 3
(b) 5
(c) 4
(d) 2

Q51. An n-channel enhancement MOSFET has V T H = 1.5 V and k = 0.5 mA/V 2 (where I D = k (V GS - V T H) 2 in saturation). For V GS = 3.5 V and V D S = 5 V , what is I D ? [Topic: MOSFET representation and characteristics | Hard]

(a) 4.0 mA
(b) 1.0 mA
(c) 2.0 mA
(d) 8.0 mA

Q52. An n-channel MOSFET with V T H = 1 V operates with V GS = 4 V . At what maximum V D S does the device remain in the triode (ohmic) region? [Topic: MOSFET representation and characteristics | Hard]

(a) Below 5 V
(b) Below 1 V
(c) Below 3 V
(d) Below 4 V

Q53. Using a digital multimeter in diode-test mode on a good silicon NPN transistor, which set of readings is expected? [Topic: Testing of diode and BJT | Hard]

(a) All three junctions read ≈ 0 V in both directions
(b) Forward B-E reads 0.3 V and B-C reads open
(c) B-E and B-C read open in both directions; C-E reads 0.7 V
(d) Forward B-E and B-C both read ≈ 0.6 – 0.7 V ; C-E reads open both ways

Q54. A diode under test shows a low resistance reading (near 0 Ω) in BOTH forward and reverse directions with an ohmmeter. What is the most likely condition? [Topic: Testing of diode and BJT | Hard]

(a) The diode is shorted
(b) The diode is leaky but usable
(c) The diode is healthy
(d) The diode is open

Q55. The reverse saturation current of a silicon diode approximately doubles for every 10°C rise. If I S = 5 nA at 25°C , what is I S at 65°C ? [Topic: Characteristics of PN junction diode | Hard]

(a) 40 nA
(b) 80 nA
(c) 20 nA
(d) 160 nA

Q56. A full-wave rectifier (center-tapped) and a half-wave rectifier both produce the same peak voltage V m . What is the ratio of their DC output voltages (full-wave to half-wave)? [Topic: Half wave and Full wave rectifier | Hard]

(a) 2 : 1
(b) 1 : 1
(c) π : 1
(d) 2 : 1

Q57. In a diode clamper circuit, a capacitor charges to the peak input voltage and shifts the DC level of the signal. For a positive clamper with input ± 10 V sine and an ideal diode, what are the output voltage limits? [Topic: Application of diodes, transistors, rectifiers and MOSFET | Hard]

(a) - 10 V to + 10 V
(b) - 20 V to 0 V
(c) 0 V to + 10 V
(d) 0 V to + 20 V

Q58. Intrinsic silicon has n i = 1.5 × 1 0 10 cm - 3 . If it is doped with donors at N D = 1.5 × 1 0 16 cm - 3 , what is the approximate equilibrium hole concentration p ? [Topic: introduction to semiconductor devices | Hard]

(a) 1.5 × 1 0 16 cm - 3
(b) 1.0 × 1 0 6 cm - 3
(c) 1.5 × 1 0 10 cm - 3
(d) 1.5 × 1 0 4 cm - 3

Q59. Below approximately 5 V , Zener diodes break down mainly by one mechanism, and above approximately 6 V by another. Which statement correctly describes their temperature coefficients? [Topic: working and operation of Zener diode | Hard]

(a) Both mechanisms have positive temperature coefficients
(b) Both mechanisms have negative temperature coefficients
(c) Zener breakdown (< 5 V) has negative tempco; avalanche (> 6 V) has positive tempco
(d) Zener breakdown has positive tempco; avalanche has negative tempco

Q60. A capacitor-filtered full-wave rectifier supplies a load drawing I L = 100 mA from a 1000 μ F capacitor at a line frequency of 50 Hz . What is the approximate peak-to-peak ripple voltage? [Topic: Application of diodes, transistors, rectifiers and MOSFET | Hard]

(a) 2.0 V
(b) 0.5 V
(c) 1.0 V
(d) 5.0 V

4.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 4 **[Subjective & Numerical Evaluation]**

1 A bridge rectifier is supplied from a 230 V, 50 Hz AC mains via a step-down transformer having a turns ratio of 10 : 1. The diodes have forward dynamic resistance $r_d = 0.5\,\Omega$ each and the load resistor is $R_L = 100\,\Omega$. Calculate:

(a) The peak secondary voltage V_m, DC load current I_{dc}, and DC output voltage V_{dc}.

(b) Rectification efficiency η and ripple factor γ without a filter.

(c) If a shunt capacitor filter of $C = 1000\,\mu$F is connected across the load, determine the resulting ripple factor and DC output voltage.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Secondary Voltage and DC Output Quantities
RMS secondary voltage: $V_{2,rms} = \frac{230}{10} = 23$ V.Peak secondary voltage: $V_m = 23\sqrt{2} \approx 32.53$ V.Two conducting diodes in series per half-cycle ($2r_d = 1.0\,\Omega$):

$$I_m = \frac{V_m}{2r_d + R_L} = \frac{32.53}{1.0 + 100} = \frac{32.53}{101} \approx 0.322 \text{ A}$$

DC load current: $I_{dc} = \frac{2I_m}{\pi} \approx 0.205$ A.DC output voltage: $V_{dc} = I_{dc}R_L = 0.205 \times 100 = 20.5$ V.**Step 2: Rectification Efficiency and Unfiltered Ripple Factor**

Rectification efficiency: $\eta = \frac{8}{\pi^2}\frac{R_L}{2r_d+R_L} \approx 80.4\%$.Ripple factor without filter: $\gamma = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} \approx 0.482$ (48.2%).**Step 3: Capacitor Filter Performance**
Ripple factor with shunt capacitor filter ($f = 50$ Hz):

$$\gamma = \frac{1}{4\sqrt{3}fCR_L} = \frac{1}{4\sqrt{3} \times 50 \times (1000 \times 10^{-6}) \times 100} \approx 0.0289 \ (2.89\%)$$

DC voltage under capacitive filtering: $V_{dc} = V_m - \frac{I_{dc}}{4fC} \approx 31.51$ V.

Examiner Insight: In bridge rectifiers, PIV per diode is V_m, which is half the $2V_m$ required in center-tapped transformers.

Chapter 5

Unit 5: Fundamentals of Filters and Operational Amplifier

5.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. A low-pass filter allows which type of signals to pass through? [Topic: low-pass filter | Easy]

(a) Only DC signals
(b) Signals within a narrow band in the middle of the spectrum, while attenuating everything above and below that band
(c) High-frequency signals
(d) Low-frequency signals

Q2. A high-pass filter attenuates signals that are: [Topic: high-pass filter | Easy]

(a) Equal to the cutoff frequency
(b) Above the cutoff frequency
(c) At all frequencies equally
(d) Below the cutoff frequency

Q3. A band-pass filter is designed to: [Topic: band-pass filter | Easy]

(a) Pass a specific range of frequencies and reject others
(b) Reject all frequencies
(c) Pass only DC voltage
(d) Pass only the lowest frequency

Q4. An ideal operational amplifier has an input impedance that is: [Topic: operational amplifier abstraction- device properties of the operational amplifier | Easy]

(a) Zero
(b) Negative
(c) Equal to 1 , k O m e g a
(d) Infinite

Q5. An ideal operational amplifier has an output impedance of: [Topic: operational amplifier abstraction- device properties of the operational amplifier | Easy]

(a) Zero
(b) 10 , k O m e g a
(c) Equal to the input impedance
(d) Infinite

Q6. The open-loop voltage gain of an ideal op-amp is: [Topic: properties of operational amplifier | Easy]

(a) Infinite
(b) Exactly 100
(c) Zero

(d) Unity

Q7. In the virtual ground concept, the voltage at the inverting input of an ideal op-amp with negative feedback is approximately: [Topic: simple op amp circuits – virtual ground concept | Easy]

(a) Equal to the supply voltage
(b) 0 , V
(c) Equal to the output voltage
(d) Infinite

Q8. The output of an inverting amplifier is: [Topic: inverting and non-inverting op-amp | Easy]

(a) Always zero
(b) In phase with the input
(c) 180° out of phase with the input
(d) 90° out of phase with the input

Q9. The voltage gain of an inverting amplifier with input resistor R 1 and feedback resistor R f is: [Topic: inverting and non-inverting op-amp | Easy]

(a) - f r a c R f R 1
(b) f r a c R 1 R f
(c) R f t im es R 1
(d) 1 + f r a c R f R 1

Q10. The voltage gain of a non-inverting amplifier is given by: [Topic: inverting and non-inverting op-amp | Easy]

(a) - f r a c R f R 1
(b) 1 + f r a c R f R 1
(c) 0
(d) f r a c R 1 R f

Q11. An op-amp summing amplifier produces an output proportional to: [Topic: op-amp as an adder and subtractor | Easy]

(a) The sum of the input voltages
(b) The difference of the input voltages
(c) The ratio of the input voltages
(d) The product of the input voltages

Q12. An op-amp subtractor circuit gives an output proportional to the: [Topic: op-amp as an adder and subtractor | Easy]

(a) Square of the input voltage
(b) Difference between two input voltages
(c) Sum of two input voltages
(d) Integral of the input voltage

Q13. An op-amp integrator produces an output that is the: [Topic: op-amp RC circuits – op-amp integrator | Easy]

(a) Inverse of the input frequency
(b) Sum of two inputs
(c) Derivative of the input signal
(d) Integral of the input signal

Q14. In a basic op-amp integrator circuit, the feedback element is a: [Topic: op-amp RC circuits – op-amp integrator | Easy]

(a) Diode
(b) Resistor
(c) Capacitor
(d) Inductor

Q15. An op-amp differentiator gives an output proportional to the: [Topic: op-amp differentiator | Easy]

(a) Integral of the input signal
(b) Rate of change of the input signal
(c) Average of the input signal
(d) Peak value of the input signal

Q16. In a basic op-amp differentiator circuit, the input element is a: [Topic: op-amp differentiator | Easy]
(a) Capacitor
(b) Resistor
(c) Transistor
(d) Inductor

Q17. An active high-pass filter differs from a passive one because it includes an: [Topic: active high pass filter | Easy]
(a) Extra ground connection
(b) Operational amplifier for gain
(c) Additional transformer
(d) External battery only

Q18. An active band-stop filter is designed to: [Topic: active band stop filter | Easy]
(a) Amplify only DC signals
(b) Pass all frequencies equally
(c) Reject a specific range of frequencies
(d) Pass only a narrow band of frequencies

Q19. Compared to passive filters, active filters generally offer: [Topic: active filter | Easy]
(a) Lower cost always
(b) No need for a power supply
(c) Gain greater than unity
(d) The ability to work only at DC

Q20. When used as a comparator, an op-amp output switches based on: [Topic: op-amp as a comparator and its application in anti-lock braking systems | Easy]
(a) Which input voltage is larger
(b) The capacitor charge time
(c) The frequency of the input
(d) The resistor tolerance in the circuit, which determines the exact threshold voltage for switching the output state

Q21. A passive RC low-pass filter has $R = 1$ k Ω and $C = 0.1$ μ F . What is its cutoff frequency? [Topic: low-pass filter | Medium]
(a) 1.00 kHz
(b) 15.9 kHz
(c) 159 Hz
(d) 1.59 kHz

Q22. In a passive RC high-pass filter, how does the output behave as the input frequency approaches zero (DC)? [Topic: high-pass filter | Medium]
(a) Output becomes infinite
(b) Output approaches zero
(c) Output doubles the input
(d) Output equals the input

Q23. A band-pass filter has a lower cutoff of 200 Hz and an upper cutoff of 2000 Hz . What is its bandwidth? [Topic: band-pass filter | Medium]
(a) 632 Hz
(b) 2200 Hz
(c) 1000 Hz

(d) 1800 Hz

Q24. An inverting amplifier uses R f = 100 k Ω and R in = 10 k Ω . What is its voltage gain? [Topic: inverting and non-inverting op-amp | Medium]

(a) - 11
(b) + 10
(c) - 10
(d) + 0.1

Q25. A non-inverting amplifier has R f = 90 k Ω and R 1 = 10 k Ω . What is its voltage gain? [Topic: inverting and non-inverting op-amp | Medium]

(a) + 9
(b) - 9
(c) + 10
(d) - 10

Q26. In an inverting op-amp circuit, why is the inverting input terminal called a virtual ground? [Topic: simple op amp circuits — virtual ground concept | Medium]

(a) It is directly wired to the ground rail
(b) It carries large current to ground
(c) It floats at the supply voltage
(d) It is held at nearly 0 V by feedback without a physical ground connection

Q27. For an ideal operational amplifier, what are the input impedance and output impedance respectively? [Topic: properties of operational amplifier | Medium]

(a) Infinite and infinite
(b) Infinite and zero
(c) Zero and zero
(d) Zero and infinite

Q28. An op-amp has an open-loop gain of 200 , 000 and a maximum output of ± 12 V . What differential input voltage drives the output to saturation? [Topic: device properties of the operational amplifier | Medium]

(a) 60 μ V
(b) 60 mV
(c) 600 μ V
(d) 6 mV

Q29. A summing amplifier has inputs 1 V and 2 V through equal input resistors R , with feedback resistor also R . What is the output voltage? [Topic: op-amp as an adder and subtractor | Medium]

(a) - 6 V
(b) + 3 V
(c) - 3 V
(d) - 1.5 V

Q30. In a basic difference amplifier with all four resistors equal, if V 1 = 3 V (non-inverting path) and V 2 = 1 V (inverting path), what is the output? [Topic: op-amp as an adder and subtractor | Medium]

(a) - 2 V
(b) - 4 V
(c) + 2 V
(d) + 4 V

Q31. An op-amp integrator is given a constant positive DC input voltage. What is the resulting output waveform? [Topic: op-amp RC circuits — op-amp integrator | Medium]

(a) A rising exponential
(b) A square wave
(c) A linearly decreasing (negative-going) ramp

(d) A constant positive level

Q32. An op-amp differentiator receives a linear ramp input. What is the output? [Topic: op-amp differentiator | Medium]

(a) A linear ramp
(b) A constant DC level
(c) An exponential decay
(d) A sine wave

Q33. An active high-pass filter uses R = 10 k Ω and C = 10 nF in its frequency-selective network. What is the cutoff frequency? [Topic: active high pass filter | Medium]

(a) 15.9 kHz
(b) 3.18 kHz
(c) 1.59 kHz
(d) 159 Hz

Q34. A first-order active band-pass filter is commonly formed by cascading which two stages? [Topic: active band pass filter | Medium]

(a) A comparator and an integrator
(b) Two identical high-pass stages
(c) Two identical low-pass stages
(d) A low-pass stage and a high-pass stage

Q35. What is the primary function of an active band-stop (notch) filter? [Topic: active band stop filter | Medium]

(a) Pass only a narrow band of frequencies
(b) Pass all frequencies above a cutoff
(c) Attenuate a specific band while passing frequencies above and below it
(d) Attenuate all frequencies equally

Q36. What is a key advantage of an active filter over an equivalent passive RC filter? [Topic: active filter | Medium]

(a) It only uses inductors and capacitors
(b) It can provide gain and avoids bulky inductors
(c) It consumes no power
(d) It works without any power supply

Q37. When an op-amp is used as a comparator with V in slightly greater than the reference voltage V re f , what is the output? [Topic: op-amp as a comparator and its application in anti-lock braking systems | Medium]

(a) Negative saturation voltage
(b) Zero volts
(c) Positive saturation voltage
(d) Exactly V re f

Q38. In an anti-lock braking system (ABS), how is an op-amp comparator typically used? [Topic: op-amp as a comparator and its application in anti-lock braking systems | Medium]

(a) To amplify the brake pedal force linearly
(b) To filter out engine noise from the audio system
(c) To integrate the vehicle's acceleration over time
(d) To compare a wheel's speed signal against a reference and detect impending lock-up

Q39. In a practical op-amp integrator, a large feedback resistor R f is placed across the capacitor mainly to: [Topic: op-amp RC circuits — op-amp integrator | Medium]

(a) Convert it into a differentiator
(b) Prevent output drift and saturation from DC offset
(c) Reduce the input impedance to zero
(d) Increase the integration speed

Q40. An op-amp has a slew rate of 1 V / μ s . What is the minimum time required for its output to swing from 0 V to 5 V ? [Topic: device properties of the operational amplifier | Medium]

(a) 1 μ s
(b) 10 μ s
(c) 5 μ s
(d) 0.2 μ s

Q41. A passive first-order RC low-pass filter has R = 10 k Ω and C = 15.9 nF . At what frequency does the output power drop to half of the maximum, and what is the phase shift there? [Topic: low-pass filter | Hard]

(a) f $\approx$ 1 kHz , phase = - 9 0 °
(b) f $\approx$ 10 kHz , phase = - 4 5 °
(c) f $\approx$ 100 Hz , phase = - 4 5 °
(d) f $\approx$ 1 kHz , phase = - 4 5 °

Q42. A first-order passive RC high-pass filter is cascaded (unbuffered) with an identical stage. Compared to a single stage, how does the overall -3 dB cutoff frequency behave? [Topic: high-pass filter | Hard]

(a) It stays exactly at the single-stage f c
(b) It shifts higher than the single-stage f c due to loading interaction
(c) It becomes independent of R and C
(d) It shifts lower than the single-stage f c

Q43. A band-pass filter has a center frequency f 0 = 2 kHz and a quality factor Q = 10 . What is its bandwidth and the approximate lower cutoff frequency? [Topic: band-pass filter | Hard]

(a) B W = 100 Hz , f L $\approx$ 1.95 kHz
(b) B W = 200 Hz , f L $\approx$ 1 kHz
(c) B W = 200 Hz , f L $\approx$ 1.9 kHz
(d) B W = 20 kHz , f L $\approx$ 1.9 kHz

Q44. In an inverting amplifier with an ideal op-amp, the inverting input is called a virtual ground. If a real op-amp has finite open-loop gain A = 1 0 5 and the closed-loop gain magnitude is 100, what is the approximate voltage at the inverting node when V o u t = 5 V ? [Topic: simple op amp circuits – virtual ground concept | Hard]

(a) 500 μ V
(b) 0 V exactly
(c) 5 mV
(d) 50 μ V

Q45. An inverting amplifier uses R in = 2 k Ω and R f = 20 k Ω . A non-inverting amplifier is built to give the same gain magnitude. What feedback and input resistor ratio must the non-inverting stage use? [Topic: inverting and non-inverting op-amp | Hard]

(a) R f / R 1 = 0.1
(b) R f / R 1 = 11
(c) R f / R 1 = 10
(d) R f / R 1 = 9

Q46. A summing amplifier has inputs V 1 = 1 V through 10 k Ω and V 2 = 2 V through 20 k Ω , with R f = 40 k Ω . What is V o u t ? [Topic: op-amp as an adder and subtractor | Hard]

(a) - 12 V
(b) - 6 V
(c) - 8 V
(d) - 4 V

Q47. An ideal op-amp integrator has R = 100 k Ω and C = 1 μ F . A constant input of 2 V is applied starting at t = 0 with zero initial capacitor charge. What is V o u t at t = 0.5 s ? [Topic: op-amp RC circuits – op-amp integrator | Hard]

(a) - 5 V

(b) - 10 V
(c) - 20 V
(d) - 2 V

Q48. A practical op-amp differentiator adds a series input resistor R 1 to the capacitor to limit high-frequency gain. If C = 10 nF , R 1 = 1 k Ω and R f = 10 k Ω , above what frequency does the circuit stop differentiating? [Topic: op-amp differentiator | Hard]

(a) ≈ 15.9 kHz
(b) ≈ 100 kHz
(c) ≈ 1.59 kHz
(d) ≈ 159 kHz

Q49. A first-order active high-pass filter (non-inverting) uses R = 15.9 k Ω , C = 10 nF , and sets a passband gain of 11. What are its cutoff frequency and passband gain in dB? [Topic: active high pass filter | Hard]

(a) f c ≈ 1 kHz , ≈ 20.8 dB
(b) f c ≈ 100 Hz , ≈ 21 dB
(c) f c ≈ 10 kHz , ≈ 20.8 dB
(d) f c ≈ 1 kHz , ≈ 11 dB

Q50. An active band-pass filter is formed by cascading a low-pass stage with f cH = 8 kHz and a high-pass stage with f c L = 500 Hz . What is the approximate center frequency (geometric mean) and bandwidth? [Topic: active band pass filter | Hard]

(a) f 0 = 4.25 kHz , B W = 7.5 kHz
(b) f 0 = 2 kHz , B W = 8.5 kHz
(c) f 0 = 2 kHz , B W = 7.5 kHz
(d) f 0 = 4 kHz , B W = 500 Hz

Q51. A band-stop (notch) filter is required to reject 50 Hz mains interference with a quality factor Q = 25 . What is the rejection bandwidth? [Topic: active band stop filter | Hard]

(a) 1250 Hz
(b) 25 Hz
(c) 0.5 Hz
(d) 2 Hz

Q52. In an ABS system, a comparator compares a wheel-speed voltage against a reference proportional to vehicle speed. Hysteresis is added using positive feedback. What is the primary benefit of this hysteresis? [Topic: op-amp as a comparator and its application in anti-lock braking systems | Hard]

(a) It converts the comparator into a linear amplifier
(b) It eliminates the need for a reference voltage
(c) It prevents rapid chattering of the output near the threshold due to noise
(d) It increases the comparator's open-loop gain

Q53. An op-amp has a gain-bandwidth product (GBW) of 10 MHz . It is configured as a non-inverting amplifier with a gain of 50. What is the approximate closed-loop bandwidth? [Topic: properties of operational amplifier | Hard]

(a) 50 kHz
(b) 500 kHz
(c) 10 MHz
(d) 200 kHz

Q54. An op-amp has a slew rate of 0.5 V / μ s . What is the maximum frequency of a sine wave with 10 V peak amplitude that can be reproduced without slew-rate distortion? [Topic: operational amplifier abstraction- device properties of the operational amplifier | Hard]

(a) ≈ 50 kHz
(b) ≈ 79.6 kHz
(c) ≈ 15.9 kHz

(d) ≈ 7.96 kHz

Q55. A difference amplifier has all four resistors equal to R . If V 1 = 3 V is applied to the non-inverting path and V 2 = 5 V to the inverting path, what is V o u t , and what is the ideal common-mode gain? [Topic: op-amp as an adder and subtractor | Hard]

(a) - 2 V , common-mode gain = 0
(b) - 2 V , common-mode gain = 1
(c) + 8 V , common-mode gain = 0
(d) + 2 V , common-mode gain = 0

Q56. A practical op-amp integrator places a resistor R f = 1 M Ω in parallel with the feedback capacitor C = 0.1 μ F , with input resistor R = 10 k Ω . What is the DC gain and why is R f needed? [Topic: op-amp RC circuits – op-amp integrator | Hard]

(a) DC gain = - 100 ; R f limits DC gain to prevent output saturation from offset/bias
(b) DC gain = - 1000 ; R f sets the unity-gain frequency
(c) DC gain = 0 ; R f provides AC coupling only
(d) DC gain = - 10 ; R f increases integration accuracy at high frequency

Q57. A Sallen-Key second-order low-pass filter is designed for a Butterworth response. What damping/quality factor and cutoff behavior characterize this maximally flat response? [Topic: active filter | Hard]

(a) Q = 0.5 , critically damped with 20 dB/decade rolloff
(b) Q = 1 , with a peaking passband and 20 dB/decade rolloff
(c) Q = 1/ 2 ≈ 0.707 , maximally flat passband with 40 dB/decade rolloff
(d) Q = 2 , with ripple in the passband and 40 dB/decade rolloff

Q58. An ideal op-amp differentiator with R f = 10 k Ω and C = 1 μ F receives a triangular input that ramps linearly from 0 to 4 V over 2 ms . What is the output voltage magnitude during this ramp? [Topic: op-amp differentiator | Hard]

(a) 40 mV
(b) 200 mV
(c) 20 mV
(d) 2 V

Q59. An inverting amplifier is designed for a gain of -20 with R f = 200 k Ω . The op-amp has an input bias current. To minimize DC output error from bias currents, what resistor should be placed at the non-inverting input? [Topic: inverting and non-inverting op-amp | Hard]

(a) 10 k Ω equal to R in
(b) 0 Ω (direct ground connection)
(c) 200 k Ω equal to R f
(d) ≈ 9.5 k Ω (parallel combination of R in and R f)

Q60. A comparator-based ABS controller detects impending wheel lock when wheel deceleration exceeds a threshold. If the wheel-speed sensor outputs a voltage proportional to speed and the circuit differentiates then compares it, what condition triggers the brake-release signal? [Topic: op-amp as a comparator and its application in anti-lock braking systems | Hard]

(a) When the raw wheel-speed voltage equals the supply voltage
(b) When the comparator operates in its linear region
(c) When the differentiated signal (deceleration) crosses the negative reference threshold
(d) When the integrated speed signal reaches zero

5.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 5 **[Subjective & Numerical Evaluation]**

1 An operational amplifier IC 741 with supply rails $V_{CC} = \pm 15$ V is utilized in the following configurations:

(a) Design an inverting summing amplifier to produce an output voltage $v_o = -(3v_1 + 2v_2 + 0.5v_3)$ using an input feedback resistor $R_f = 60$ kΩ.

(b) In an active first-order low-pass Butterworth filter with a passband voltage gain of 1.586 and cutoff frequency $f_c = 2.5$ kHz, calculate the component values of R and C (choosing $C = 10$ nF) and feedback resistors R_1, R_f.

(c) Formulate how an op-amp comparator serves as the threshold detector in an automotive Anti-Lock Braking System (ABS) wheel-slip detection unit.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Inverting Summing Amplifier Design
Output equation: $v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$.With $R_f = 60$ kΩ:

$$R_1 = \frac{60\ \text{k}\Omega}{3} = 20\ \text{k}\Omega, \quad R_2 = \frac{60\ \text{k}\Omega}{2} = 30\ \text{k}\Omega, \quad R_3 = \frac{60\ \text{k}\Omega}{0.5} = 120\ \text{k}\Omega$$

Step 2: Active First-Order Low-Pass Filter Design
Given $f_c = 2.5$ kHz and $C = 10$ nF:

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \times 2500 \times 10 \times 10^{-9}} \approx 6.37\ \text{k}\Omega$$

For non-inverting passband gain $A_v = 1 + \frac{R_f}{R_1} = 1.586$:

$$\frac{R_f}{R_1} = 0.586 \implies \text{Selecting } R_1 = 10\ \text{k}\Omega \implies R_f = 5.86\ \text{k}\Omega$$

Step 3: Anti-lock Braking System (ABS) Comparator Architecture
In modern automotive ABS, variable reluctance wheel-speed sensors produce a pulsed AC voltage proportional to wheel speed ω_w.

- Inverting terminal receives conditioned sensor signal v_{sensor}.
- Non-inverting terminal receives reference threshold V_{ref} from the Electronic Control Unit.
- When wheel slip occurs (impending lockup), sensor voltage drops below V_{ref}, causing comparator output to jump to $+V_{sat}$ (logic HIGH).
- The ECU pulses solenoid valves to release and modulate brake caliper pressure, preserving vehicle steering control.

Key Takeaway: Stating virtual ground ($V^+ \approx V^-$) and zero input bias current ($I_{in} \approx 0$) is vital for full marks in op-amp proofs.

Chapter 6

Unit 6: Embedded System and Its Application

6.1 Section A: Multiple Choice Question Practice (60 Curated Problems)

Q1. What is an embedded system? [Topic: introduction to embedded system | Easy]

(a) A type of operating system for servers
(b) A general-purpose computer used for office work
(c) A dedicated computer system designed to perform a specific task
(d) A software used only for gaming

Q2. Which of the following is a common example of an embedded system? [Topic: introduction to embedded system | Easy]

(a) A spreadsheet application
(b) A desktop PC
(c) A cloud data center
(d) A washing machine controller

Q3. Which component acts as the 'brain' of an embedded system? [Topic: introduction to embedded system | Easy]

(a) Plastic casing
(b) Processor (microcontroller/microprocessor)
(c) Display screen
(d) Power cable

Q4. Embedded systems are generally designed to be: [Topic: introduction to embedded system | Easy]

(a) Compact, reliable, and low power
(b) Large and expensive
(c) Only used in laboratories
(d) Incapable of real-time operation

Q5. Which device integrates CPU, memory, and I/O ports on a single chip? [Topic: comparison of microprocessor and micro-controller | Easy]

(a) Capacitor
(b) Microcontroller
(c) Microprocessor
(d) Transistor

Q6. A microprocessor typically requires which of the following to function? [Topic: comparison of microprocessor and micro-controller | Easy]

(a) External memory and peripherals
(b) Only a display

(c) No additional components at all
(d) Only a battery

Q7. Which is generally more suitable for small, dedicated control applications? [Topic: comparison of microprocessor and micro-controller | Easy]
(a) Supercomputer
(b) Microprocessor
(c) Mainframe
(d) Microcontroller

Q8. Compared to a microcontroller, a microprocessor is usually used in: [Topic: comparison of microprocessor and micro-controller | Easy]
(a) Only in sensors
(b) Simple toys only
(c) High-performance computing systems like PCs
(d) Only in LED lights

Q9. Which generally consumes less power for embedded tasks? [Topic: comparison of microprocessor and micro-controller | Easy]
(a) Both consume equal very high power
(b) Microprocessor
(c) Neither consumes any power
(d) Microcontroller

Q10. Arduino is best described as a/an: [Topic: introduction of arduino and sensors | Easy]
(a) Cloud storage service
(b) High-voltage transformer
(c) Open-source microcontroller-based development board
(d) Operating system for laptops

Q11. What is the main function of a sensor? [Topic: introduction of arduino and sensors | Easy]
(a) To display video output
(b) To supply high voltage
(c) To convert a physical quantity into an electrical signal
(d) To store large amounts of data permanently

Q12. Which of the following is an example of a temperature sensor? [Topic: introduction of arduino and sensors | Easy]
(a) LM35
(b) LED
(c) Resistor
(d) Relay

Q13. The programming environment used to write code for Arduino is called the: [Topic: introduction of arduino and sensors | Easy]
(a) Excel Macro
(b) Photoshop
(c) Arduino IDE
(d) Microsoft Word

Q14. Arduino boards are commonly programmed using a language based on: [Topic: introduction of arduino and sensors | Easy]
(a) Assembly only
(b) SQL
(c) C/C++
(d) HTML

Q15. What does IoT stand for? [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]

(a) Input of Technology
(b) Internet of Transistors
(c) Integration of Tools
(d) Internet of Things

Q16. In healthcare, IoT devices are commonly used to: [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]
(a) Remotely monitor patient vital signs
(b) Replace all doctors permanently
(c) Generate electricity for hospitals
(d) Store printed paper records

Q17. In agriculture, IoT sensors are mainly used to monitor: [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]
(a) Internet browsing history
(b) Soil moisture and crop conditions
(c) Television channels
(d) Mobile phone games

Q18. What is the main role of cloud computing in IoT systems? [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]
(a) Storing and processing data from connected devices
(b) Manufacturing microcontrollers
(c) Supplying electrical power to devices
(d) Generating physical sensors

Q19. A smart grid primarily uses IoT and sensors to: [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]
(a) Monitor and manage electricity distribution efficiently
(b) Stop electricity usage completely
(c) Produce coal for power plants
(d) Replace all power lines with batteries

Q20. Condition monitoring of plant machinery using IoT helps mainly to: [Topic: role of IOT and cloud computing in condition monitoring of plant processes, health care, agriculture, manufacturing, automobiles and smart grid | Easy]
(a) Detect faults early and reduce downtime
(b) Remove the need for electricity
(c) Make machines heavier
(d) Increase the size of machines

Q21. A digital washing machine controller is dedicated to running only wash cycles and cannot be reprogrammed by the user for other tasks. Which characteristic of embedded systems does this best illustrate? [Topic: introduction to embedded system | Medium]
(a) Application-specific and dedicated functionality
(b) Unlimited reconfigurable software
(c) General-purpose computing capability
(d) High-latency batch processing

Q22. Which combination of constraints is most typical when designing a battery-powered embedded device such as a fitness band? [Topic: introduction to embedded system | Medium]
(a) No timing constraints and large size
(b) Low power consumption and real-time response
(c) Maximum cost and manual operation

(d) High power draw and unlimited memory

Q23. A real-time embedded system in an airbag controller must respond within a strict deadline. Missing this deadline causes system failure. This is an example of which type of system? [Topic: introduction to embedded system | Medium]

(a) Batch processing system
(b) Hard real-time system
(c) Soft real-time system
(d) Non-real-time system

Q24. An engineer needs a single chip containing CPU, RAM, ROM, and I/O ports for a low-cost temperature logger. Which device is the most suitable choice? [Topic: comparison of microprocessor and micro-controller | Medium]

(a) Microcontroller
(b) Discrete logic gates
(c) Microprocessor
(d) External DRAM module

Q25. Why does a microprocessor-based system generally require more external components than a microcontroller-based system? [Topic: comparison of microprocessor and micro-controller | Medium]

(a) The microprocessor has slower clock speeds
(b) The microprocessor consumes no power
(c) Memory and I/O are not integrated on the microprocessor chip
(d) The microprocessor cannot execute arithmetic

Q26. Which statement best compares typical use cases of microprocessors and microcontrollers? [Topic: comparison of microprocessor and micro-controller | Medium]

(a) Microprocessors suit dedicated control; microcontrollers suit general computing
(b) Both are unable to interface with sensors
(c) Both are used only in desktop computers
(d) Microprocessors suit computation-heavy general systems; microcontrollers suit dedicated control tasks

Q27. A microcontroller is often said to follow a Harvard architecture while many microprocessors use von Neumann architecture. What is the key implication of Harvard architecture? [Topic: comparison of microprocessor and micro-controller | Medium]

(a) Elimination of the CPU
(b) Separate buses for program and data memory
(c) No ability to store programs
(d) A single shared bus for all memory

Q28. On an Arduino Uno, which function is used to read a varying analog voltage from a sensor such as an LDR? [Topic: introduction of arduino and sensors | Medium]

(a) pinMode()
(b) analogRead()
(c) digitalWrite()
(d) Serial.print()

Q29. The Arduino Uno's analog-to-digital converter has 10-bit resolution. How many discrete levels can it represent? [Topic: introduction of arduino and sensors | Medium]

(a) 512
(b) 4096
(c) 1024
(d) 256

Q30. A designer wants to measure distance to an obstacle for an automatic parking system. Which sensor is most appropriate? [Topic: introduction of arduino and sensors | Medium]

(a) LDR (light sensor)
(b) Thermistor

(c) Hall-effect switch
(d) Ultrasonic sensor

Q31. Which classification best describes a thermocouple used with an Arduino to measure temperature? [Topic: introduction of arduino and sensors | Medium]
(a) Analog sensor producing a continuous voltage
(b) Actuator converting electrical to motion
(c) Communication module
(d) Digital sensor producing only 0 or 1

Q32. In an Arduino sketch, which function runs continuously and repeatedly after startup to handle ongoing sensor polling? [Topic: introduction of arduino and sensors | Medium]
(a) setup()
(b) loop()
(c) main()
(d) init()

Q33. In an IoT-based plant condition-monitoring system, vibration sensors on a motor send data to the cloud, which detects a rising trend before failure. This capability is best described as: [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Medium]
(a) Random fault guessing
(b) Reactive repair after breakdown
(c) Predictive maintenance
(d) Manual periodic inspection only

Q34. Why is cloud computing advantageous for storing and analyzing data from thousands of plant sensors? [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Medium]
(a) It offers scalable storage and processing on demand
(b) It permanently stops all machine faults
(c) It eliminates the need for any sensors
(d) It requires no network connectivity

Q35. An IoT monitoring system needs instant local reaction to a dangerous pressure spike without waiting for the cloud. Which approach addresses this requirement? [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Medium]
(a) Sending all data only to a distant cloud
(b) Monthly manual data upload
(c) Disabling the sensors
(d) Edge computing near the sensor

Q36. A wearable IoT device continuously transmits a patient's heart rate to doctors for remote observation. This application is best known as: [Topic: health care | Medium]
(a) Remote patient monitoring
(b) In-person surgery
(c) Hospital catering
(d) Pharmacy billing

Q37. A smart-farming IoT system uses soil-moisture sensors to turn on irrigation only when moisture drops below a threshold. What is the primary benefit? [Topic: agriculture | Medium]
(a) Increased water wastage
(b) Removal of all crops
(c) Efficient water usage
(d) Higher fuel consumption

Q38. In a smart manufacturing line, IoT sensors feed real-time production data to a dashboard that automatically flags defective units. This is a core concept of: [Topic: manufacturing | Medium]
(a) Industry 4.0
(b) Manual assembly only

(c) Paper-based record keeping
(d) Pre-electricity production

Q39. A connected car uploads engine diagnostics to the cloud, which alerts the driver about an impending component failure. This feature primarily relies on: [Topic: automobiles | Medium]
(a) IoT telematics and cloud analytics
(b) A standalone offline odometer
(c) Printed service manuals
(d) Mechanical gear ratios only

Q40. A smart grid uses IoT-enabled smart meters that allow two-way communication between consumers and the utility. What is a key advantage of this over traditional meters? [Topic: smart grid | Medium]
(a) Reading only once per year manually
(b) Inability to measure consumption
(c) No communication with the utility
(d) Real-time demand response and dynamic load management

Q41. A real-time embedded system must guarantee a response to a sensor interrupt within 50 μ s . The CPU runs at 20 MHz and the interrupt latency plus the ISR consumes 600 clock cycles. By what minimum factor must the clock be increased to safely meet a stricter deadline of 20 μ s ? [Topic: introduction to embedded system | Hard]
(a) 1.5 $\times$
(b) 2.5 $\times$
(c) 3 $\times$
(d) 5 $\times$

Q42. Which characteristic most sharply distinguishes a hard real-time embedded system from a soft real-time embedded system? [Topic: introduction to embedded system | Hard]
(a) It always uses a faster processor than soft real-time systems
(b) It never uses interrupts for timing control
(c) Missing a deadline causes total system failure, not just degraded quality
(d) It stores more data in non-volatile memory

Q43. A design requires on-chip RAM, ROM, timers, and ADC in a single IC for a low-cost washing machine controller. Choosing a microprocessor instead of a microcontroller would MOST directly cause which penalty? [Topic: comparison of microprocessor and micro-controller | Hard]
(a) Increased board area, cost, and power due to external peripheral chips
(b) Loss of the ability to run assembly language
(c) Reduced instruction set width limiting arithmetic
(d) Inability to execute any interrupt service routine

Q44. The 8051 microcontroller uses a Harvard architecture while a typical general-purpose microprocessor uses von Neumann. What is the key performance consequence of this difference? [Topic: comparison of microprocessor and micro-controller | Hard]
(a) Simultaneous instruction fetch and data access improve throughput
(b) Interrupts are completely eliminated
(c) Program memory can never be larger than data memory
(d) Only one addressing mode becomes available

Q45. A system must perform heavy floating-point image processing with a large external DRAM and run a full OS. Which selection and reasoning is correct? [Topic: comparison of microprocessor and micro-controller | Hard]
(a) Microcontroller, because its on-chip RAM handles large images
(b) Microprocessor, because it scales with large external memory and OS support
(c) Microprocessor, because it cannot interface external memory
(d) Microcontroller, because it has a built-in FPU superior to any microprocessor

Q46. An Arduino Uno (ATmega328P) uses a 10 -bit ADC with a 5 V reference. A sensor outputs

2.5 V . What raw ADC value and resolution per step apply? [Topic: introduction of arduino and sensors | Hard]

(a) Value ≈ 512 , step ≈ 9.77 mV
(b) Value ≈ 512 , step ≈ 4.88 mV
(c) Value ≈ 1023 , step ≈ 4.88 mV
(d) Value ≈ 256 , step ≈ 4.88 mV

Q47. A designer connects a 3.3 V logic sensor's output to an Arduino Uno (5 V logic) input and reads correct HIGH values, but connecting the Uno's 5 V output to the sensor's input risks damage. What is the correct explanation? [Topic: introduction of arduino and sensors | Hard]

(a) Both directions are always safe without level shifting
(b) 3.3 V exceeds the Uno's HIGH threshold, but 5 V exceeds the sensor's max input rating
(c) The Uno interprets 3.3 V as LOW always
(d) The sensor cannot output any logic HIGH at 3.3 V

Q48. A resistive sensor is read via a voltage divider with a fixed 10 k Ω resistor and 5 V supply. For best ADC sensitivity near the sensor's nominal resistance R s = 10 k Ω , which statement is correct? [Topic: introduction of arduino and sensors | Hard]

(a) Output voltage is independent of R s
(b) The ADC cannot resolve any change at this point
(c) Sensitivity is maximized because the divider output is near mid-scale (2.5 V)
(d) Sensitivity is lowest because output saturates at 5 V

Q49. Why does the Arduino delay() function create problems in a system that must simultaneously poll three sensors at different rates? [Topic: introduction of arduino and sensors | Hard]

(a) It increases the ADC resolution unpredictably
(b) It resets the microcontroller after each call
(c) It disables all GPIO pins permanently
(d) It blocks the CPU, preventing other tasks during the delay interval

Q50. In a plant condition-monitoring system, vibration sensors sample at 10 kHz . To detect a fault signature at 4 kHz without aliasing, which sampling and processing choice is correct? [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Hard]

(a) 10 kHz is below Nyquist for 4 kHz , so detection fails
(b) Sampling frequency is irrelevant to aliasing
(c) A 4 kHz signal requires at least 4 kHz sampling only
(d) 10 kHz satisfies Nyquist ($> 2 \times 4$ kHz), so 4 kHz is detectable

Q51. A plant uses edge computing before sending data to the cloud. Which scenario BEST justifies processing at the edge rather than purely in the cloud? [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Hard]

(a) Cross-site statistical benchmarking of all plants
(b) Monthly aggregate report generation
(c) Long-term trend archival over several years
(d) Low-latency shutdown decisions needed within milliseconds on local vibration spikes

Q52. An IoT gateway transmits 500 sensor readings per second, each 20 bytes , over a link. What minimum raw throughput is required, ignoring protocol overhead? [Topic: role of IOT and cloud computing in condition monitoring of plant processes | Hard]

(a) 40 kbps
(b) 80 kbps
(c) 160 kbps
(d) 10 kbps

Q53. A wearable IoT ECG device must prioritize detecting life-threatening arrhythmia locally while uploading full waveforms to the cloud. Which architecture best balances reliability and bandwidth? [Topic: health care | Hard]

(a) Run arrhythmia detection on-device; stream compressed waveforms to cloud for review

(b) Send every raw sample to cloud and wait for cloud diagnosis before alerting
(c) Store all data on-device only and never transmit
(d) Perform all detection in the cloud with no local processing

Q54. In a cloud-connected patient monitoring system, which design flaw most directly violates patient data security requirements? [Topic: health care | Hard]
(a) Using redundant cloud storage regions
(b) Buffering readings locally before upload
(c) Transmitting physiological data over unencrypted HTTP
(d) Compressing waveforms before transmission

Q55. A smart irrigation system uses soil-moisture sensors and a cloud model. During a multi-day network outage, which edge-level fallback best preserves crop health? [Topic: agriculture | Hard]
(a) System irrigates continuously at maximum flow
(b) Local controller irrigates using threshold rules on cached sensor data
(c) System deletes sensor data to save memory
(d) System halts all irrigation until cloud reconnects

Q56. A field deploys 200 LoRa soil sensors each reporting once every 10 minutes . What is the average aggregate message rate the gateway must handle? [Topic: agriculture | Hard]
(a) ≈ 0.33 messages/s
(b) ≈ 20 messages/s
(c) ≈ 3.3 messages/s
(d) ≈ 2 messages/s

Q57. In a smart factory, predictive maintenance uses IoT vibration data and cloud ML. What is the primary advantage over traditional scheduled (time-based) maintenance? [Topic: manufacturing | Hard]
(a) Maintenance is triggered by actual equipment condition, reducing unplanned downtime
(b) It removes the need for cloud or network connectivity
(c) It guarantees machines never fail under any circumstance
(d) It eliminates the need for any sensors on machines

Q58. A manufacturing line's IoT system shows rising false-alarm rates after adding more sensors. Which root cause is most consistent with this symptom? [Topic: manufacturing | Hard]
(a) Removal of the cloud analytics layer
(b) Reduced total data volume from the line
(c) Fixed thresholds not adapted to increased sensor noise and correlation
(d) Lower sampling frequency on all channels

Q59. In a connected vehicle, which function MUST remain on an embedded ECU rather than depend on cloud computing, and why? [Topic: automobiles | Hard]
(a) Over-the-air firmware distribution, due to version management
(b) Infotainment music streaming, due to large data size
(c) Anti-lock braking control, due to millisecond-level safety-critical latency
(d) Fleet-wide fuel-economy benchmarking, due to data aggregation

Q60. A smart grid uses IoT smart meters for demand response. During a sudden generation shortfall, which cloud+edge mechanism most effectively prevents a cascading blackout? [Topic: smart grid | Hard]
(a) Disabling all smart meters until supply recovers
(b) Increasing data logging resolution only
(c) Edge-level automatic load shedding guided by cloud demand-response signals
(d) Waiting for monthly cloud billing reports to adjust tariffs

6.2 Section B: Comprehensive Analytical Problems with Step-by-Step Solutions

Analytical Problem 6 **[Subjective & Numerical Evaluation]**

1 An Arduino Uno microcontroller development system based on the ATmega328P is deployed to monitor agricultural conditions using an analog temperature sensor (LM35) and a soil moisture sensor.

(a) Contrast the architecture of the ATmega328P microcontroller against an Intel Core i7 microprocessor across memory organization, power consumption, execution determinism, and peripheral integration.

(b) The ATmega328P features a 10-bit Successive Approximation ADC operating with reference voltage $V_{ref} = 5.0$ V. Compute the quantization step size. If the LM35 produces 10 mV/°C and reads 35.0°C, determine the analog voltage and the digital ADC code recorded.

(c) Outline the end-to-end cloud IoT architecture for telemetry streaming from this sensor node to a remote cloud analytics dashboard via MQTT/Wi-Fi.

Step-by-Step Rigorous Mathematical Solution:

Step 1: Architectural Comparison (Microcontroller vs Microprocessor)

Feature	ATmega328P (Microcontroller)	Intel Core i7 (Microprocessor)
Architecture	Harvard (Separate code and data buses)	Modified Harvard / Unified cache
Hardware On-Chip	CPU, 32KB Flash, 2KB SRAM, ADC, Timers, UART on single chip	CPU cores, cache only; requires external RAM, GPU, Chipset
Power	Very low (< 50 mW)	High (65–125 W)
Determinism	Hard real-time cycle-accurate execution	Non-deterministic OS latency
Typical Domain	Dedicated sensing and actuator control	General-purpose computing systems

Step 2: ADC Quantization and Sensor Signal Digital Conversion
A 10-bit ADC provides $2^{10} = 1024$ discrete digital levels (0 to 1023).The quantization step size (V_{LSB}) is:

$$V_{LSB} = \frac{V_{ref}}{1024} = \frac{5.0 \text{ V}}{1024} \approx 4.8828 \text{ mV}$$

At $T = 35.0°\text{C}$, the LM35 output voltage is $V_{out} = 35.0 \times 10 \text{ mV} = 350.0 \text{ mV} = 0.350 \text{ V}$.The digital ADC count is:

$$D = \left\lfloor \frac{0.350}{5.0} \times 1024 \right\rfloor = \lfloor 71.68 \rfloor = 71 \text{ or } 72$$

Step 3: End-to-End Cloud IoT Telemetry Architecture

1. **Perception Layer**: Sensors convert physical quantities into analog voltages.
2. **Edge Layer**: ATmega328P digitizes and serializes measurements into JSON packets.
3. **Network Layer**: Wi-Fi module transmits packets using MQTT over TLS to the cloud broker.
4. **Application Layer**: Cloud service ingests, stores in time-series DB, and displays real-time telemetry.

Implementation Note: Using the internal 1.1 V reference increases LM35 sensitivity to ≈ 1.074 mV/step.